

Useful Tables

Appendix

A

Appendix Outline

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Table A-1

Standard SI Prefixes*†

Name	Symbol	Factor
exa	E	1 000 000 000 000 000 000 = 10^{18}
peta	P	1 000 000 000 000 000 = 10^{15}
tera	T	1 000 000 000 000 = 10^{12}
giga	G	1 000 000 000 = 10^9
mega	M	1 000 000 = 10^6
kilo	k	1 000 = 10^3
hecto‡	h	100 = 10^2
deka‡	da	10 = 10^1
deci‡	d	0.1 = 10^{-1}
centi‡	c	0.01 = 10^{-2}
milli	m	0.001 = 10^{-3}
micro	μ	0.000 001 = 10^{-6}
nano	n	0.000 000 001 = 10^{-9}
pico	p	0.000 000 000 001 = 10^{-12}
femto	f	0.000 000 000 000 001 = 10^{-15}
atto	a	0.000 000 000 000 000 001 = 10^{-18}

*If possible use multiple and submultiple prefixes in steps of 1000.

†Spaces are used in SI instead of commas to group numbers to avoid confusion with the practice in some European countries of using commas for decimal points.

‡Not recommended but sometimes encountered.

Table A-2Conversion Factors A to Convert Input X to Output Y Using the Formula $Y = AX^*$

Multiply Input X	By Factor A	To Get Output Y	Multiply Input X	By Factor A	To Get Output Y
British thermal unit, Btu	1055	joule, J	mile/hour, mi/h	1.61	kilometer/hour, km/h
Btu/second, Btu/s	1.05	kilowatt, kW	mile/hour, mi/h	0.447	meter/second, m/s
calorie	4.19	joule, J	moment of inertia, lbm · ft ²	0.0421	kilogram-meter ² , kg · m ²
centimeter of mercury (0°C)	1.333	kilopascal, kPa	moment of inertia, lbm · in ²	293	kilogram-millimeter ² , kg · mm ²
centipoise, cP	0.001	pascal-second, Pa · s	moment of section (second moment of area), in ⁴	41.6	centimeter ⁴ , cm ⁴
degree (angle)	0.0174	radian, rad	ounce-force, oz	0.278	newton, N
foot, ft	0.305	meter, m	ounce-mass	0.0311	kilogram, kg
foot ² , ft ²	0.0929	meter ² , m ²	pound, lbf [†]	4.45	newton, N
foot/minute, ft/min	0.0051	meter/second, m/s	pound-foot, lbf · ft	1.36	newton-meter, N · m
foot-pound, ft · lbf	1.35	joule, J	pound/foot ² , lbf/ft ²	47.9	pascal, Pa
foot-pound/second, ft · lbf/s	1.35	watt, W	pound-inch, lbf · in	0.113	joule, J
foot/second, ft/s	0.305	meter/second, m/s	pound-inch, lbf · in	0.113	newton-meter, N · m
gallon (U.S.), gal	3.785	liter, L	pound/inch, lbf/in	175	newton/meter, N/m
horsepower, hp	0.746	kilowatt, kW	pound/inch ² , psi (lbf/in ²)	6.89	kilopascal, kPa
inch, in	0.0254	meter, m	pound-mass, lbm	0.454	kilogram, kg
inch, in	25.4	millimeter, mm	pound-mass/second, lbm/s	0.454	kilogram/second, kg/s
inch ² , in ²	645	millimeter ² , mm ²	quart (U.S. liquid), qt	946	milliliter, mL
inch of mercury (32°F)	3.386	kilopascal, kPa	section modulus, in ³	16.4	centimeter ³ , cm ³
kilopound, kip	4.45	kilonewton, kN	slug	14.6	kilogram, kg
kilopound/inch ² , kpsi (ksi)	6.89	megapascal, MPa (N/mm ²)	ton (short 2000 lbm)	907	kilogram, kg
mass, lbf · s ² /in	175	kilogram, kg	yard, yd	0.914	meter, m
mile, mi	1.610	kilometer, km			

*Approximate.

[†]The U.S. Customary system unit of the pound-force is often abbreviated as lbf to distinguish it from the pound-mass, which is abbreviated as lbm.

Table A-3

Optional SI Units for
Bending Stress

$\sigma = Mc/I$, Torsion Stress

$\tau = Tr/J$, Axial Stress σ

$= F/A$, and Direct

Shear Stress

$\tau = F/A$

Bending and Torsion				Axial and Direct Shear		
M, T	I, J	c, r	σ, τ	F	A	σ, τ
$\text{N} \cdot \text{m}^*$	m^4	m	Pa	N^*	m^2	Pa
$\text{N} \cdot \text{m}$	cm^4	cm	$\text{MPa (N/mm}^2\text{)}$	$\text{N}^\dagger$	mm^2	$\text{MPa (N/mm}^2\text{)}$
$\text{N} \cdot \text{m}^\dagger$	mm^4	mm	GPa	kN	m^2	kPa
$\text{kN} \cdot \text{m}$	cm^4	cm	GPa	$\text{kN}^\dagger$	mm^2	GPa
$\text{N} \cdot \text{mm}^\dagger$	mm^4	mm	$\text{MPa (N/mm}^2\text{)}$			

*Basic relation.

†Often preferred.

Table A-4

Optional SI Units for
Bending Deflection

$y = f(Fl^3/EI)$ or

$y = f(wl^4/EI)$ and

Torsional Deflection

$\theta = Tl/GJ$

Bending Deflection						Torsional Deflection			
F, wl	l	I	E	y	T	l	J	G	θ
N^*	m	m^4	Pa	m	$\text{N} \cdot \text{m}^*$	m	m^4	Pa	rad
$\text{kN}^\dagger$	mm	mm^4	GPa	mm	$\text{N} \cdot \text{m}^\dagger$	mm	mm^4	GPa	rad
kN	m	m^4	GPa	μm	$\text{N} \cdot \text{mm}$	mm	mm^4	$\text{MPa (N/mm}^2\text{)}$	rad
N	mm	mm^4	kPa	m	$\text{N} \cdot \text{m}$	cm	cm^4	$\text{MPa (N/mm}^2\text{)}$	rad

*Basic relation.

†Often preferred.

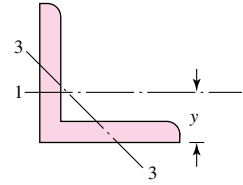
Table A-5

Physical Constants of Materials

Material	Modulus of Elasticity E		Modulus of Rigidity G		Poisson's Ratio ν	Unit Weight w		
	Mpsi	GPa	Mpsi	GPa		lbf/in³	lbf/ft³	kN/m³
Aluminum (all alloys)	10.4	71.7	3.9	26.9	0.333	0.098	169	26.6
Beryllium copper	18.0	124.0	7.0	48.3	0.285	0.297	513	80.6
Brass	15.4	106.0	5.82	40.1	0.324	0.309	534	83.8
Carbon steel	30.0	207.0	11.5	79.3	0.292	0.282	487	76.5
Cast iron (gray)	14.5	100.0	6.0	41.4	0.211	0.260	450	70.6
Copper	17.2	119.0	6.49	44.7	0.326	0.322	556	87.3
Douglas fir	1.6	11.0	0.6	4.1	0.33	0.016	28	4.3
Glass	6.7	46.2	2.7	18.6	0.245	0.094	162	25.4
Inconel	31.0	214.0	11.0	75.8	0.290	0.307	530	83.3
Lead	5.3	36.5	1.9	13.1	0.425	0.411	710	111.5
Magnesium	6.5	44.8	2.4	16.5	0.350	0.065	112	17.6
Molybdenum	48.0	331.0	17.0	117.0	0.307	0.368	636	100.0
Monel metal	26.0	179.0	9.5	65.5	0.320	0.319	551	86.6
Nickel silver	18.5	127.0	7.0	48.3	0.322	0.316	546	85.8
Nickel steel	30.0	207.0	11.5	79.3	0.291	0.280	484	76.0
Phosphor bronze	16.1	111.0	6.0	41.4	0.349	0.295	510	80.1
Stainless steel (18-8)	27.6	190.0	10.6	73.1	0.305	0.280	484	76.0
Titanium alloys	16.5	114.0	6.2	42.4	0.340	0.160	276	43.4

Table A-6Properties of Structural-
Steel Angles*†

w = weight per foot, lbf/ft
 m = mass per meter, kg/m
 A = area, in² (cm²)
 I = second moment of area, in⁴ (cm⁴)
 k = radius of gyration, in (cm)
 y = centroidal distance, in (cm)
 Z = section modulus, in³, (cm³)



Size, in	w	A	I_{1-1}	k_{1-1}	Z_{1-1}	y	k_{3-3}
$1 \times 1 \times \frac{1}{8}$	0.80	0.234	0.021	0.298	0.029	0.290	0.191
$\times \frac{1}{4}$	1.49	0.437	0.036	0.287	0.054	0.336	0.193
$1\frac{1}{2} \times 1\frac{1}{2} \times \frac{1}{8}$	1.23	0.36	0.074	0.45	0.068	0.41	0.29
$\times \frac{1}{4}$	2.34	0.69	0.135	0.44	0.130	0.46	0.29
$2 \times 2 \times \frac{1}{8}$	1.65	0.484	0.190	0.626	0.131	0.546	0.398
$\times \frac{1}{4}$	3.19	0.938	0.348	0.609	0.247	0.592	0.391
$\times \frac{3}{8}$	4.7	1.36	0.479	0.594	0.351	0.636	0.389
$2\frac{1}{2} \times 2\frac{1}{2} \times \frac{1}{4}$	4.1	1.19	0.703	0.769	0.394	0.717	0.491
$\times \frac{3}{8}$	5.9	1.73	0.984	0.753	0.566	0.762	0.487
$3 \times 3 \times \frac{1}{4}$	4.9	1.44	1.24	0.930	0.577	0.842	0.592
$\times \frac{3}{8}$	7.2	2.11	1.76	0.913	0.833	0.888	0.587
$\times \frac{1}{2}$	9.4	2.75	2.22	0.898	1.07	0.932	0.584
$3\frac{1}{2} \times 3\frac{1}{2} \times \frac{1}{4}$	5.8	1.69	2.01	1.09	0.794	0.968	0.694
$\times \frac{3}{8}$	8.5	2.48	2.87	1.07	1.15	1.01	0.687
$\times \frac{1}{2}$	11.1	3.25	3.64	1.06	1.49	1.06	0.683
$4 \times 4 \times \frac{1}{4}$	6.6	1.94	3.04	1.25	1.05	1.09	0.795
$\times \frac{3}{8}$	9.8	2.86	4.36	1.23	1.52	1.14	0.788
$\times \frac{1}{2}$	12.8	3.75	5.56	1.22	1.97	1.18	0.782
$\times \frac{5}{8}$	15.7	4.61	6.66	1.20	2.40	1.23	0.779
$6 \times 6 \times \frac{3}{8}$	14.9	4.36	15.4	1.88	3.53	1.64	1.19
$\times \frac{1}{2}$	19.6	5.75	19.9	1.86	4.61	1.68	1.18
$\times \frac{5}{8}$	24.2	7.11	24.2	1.84	5.66	1.73	1.18
$\times \frac{3}{4}$	28.7	8.44	28.2	1.83	6.66	1.78	1.17

Table A-6

Properties of Structural-
Steel Angles*†
(Continued)

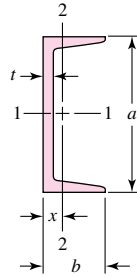
Size, mm	<i>m</i>	<i>A</i>	<i>I</i> ₁₋₁	<i>k</i> ₁₋₁	<i>Z</i> ₁₋₁	<i>y</i>	<i>k</i> ₃₋₃
25 × 25 × 3	1.11	1.42	0.80	0.75	0.45	0.72	0.48
× 4	1.45	1.85	1.01	0.74	0.58	0.76	0.48
× 5	1.77	2.26	1.20	0.73	0.71	0.80	0.48
40 × 40 × 4	2.42	3.08	4.47	1.21	1.55	1.12	0.78
× 5	2.97	3.79	5.43	1.20	1.91	1.16	0.77
× 6	3.52	4.48	6.31	1.19	2.26	1.20	0.77
50 × 50 × 5	3.77	4.80	11.0	1.51	3.05	1.40	0.97
× 6	4.47	5.59	12.8	1.50	3.61	1.45	0.97
× 8	5.82	7.41	16.3	1.48	4.68	1.52	0.96
60 × 60 × 5	4.57	5.82	19.4	1.82	4.45	1.64	1.17
× 6	5.42	6.91	22.8	1.82	5.29	1.69	1.17
× 8	7.09	9.03	29.2	1.80	6.89	1.77	1.16
× 10	8.69	11.1	34.9	1.78	8.41	1.85	1.16
80 × 80 × 6	7.34	9.35	55.8	2.44	9.57	2.17	1.57
× 8	9.63	12.3	72.2	2.43	12.6	2.26	1.56
× 10	11.9	15.1	87.5	2.41	15.4	2.34	1.55
100 × 100 × 8	12.2	15.5	145	3.06	19.9	2.74	1.96
× 12	17.8	22.7	207	3.02	29.1	2.90	1.94
× 15	21.9	27.9	249	2.98	35.6	3.02	1.93
150 × 150 × 10	23.0	29.3	624	4.62	56.9	4.03	2.97
× 12	27.3	34.8	737	4.60	67.7	4.12	2.95
× 15	33.8	43.0	898	4.57	83.5	4.25	2.93
× 18	40.1	51.0	1050	4.54	98.7	4.37	2.92

*Metric sizes also available in sizes of 45, 70, 90, 120, and 200 mm.

†These sizes are also available in aluminum alloy.

Table A-7

Properties of Structural-Steel Channels*

 a, b = size, in (mm) w = weight per foot, lbf/ft m = mass per meter, kg/m t = web thickness, in (mm) A = area, in² (cm²) I = second moment of area, in⁴ (cm⁴) k = radius of gyration, in (cm) x = centroidal distance, in (cm) Z = section modulus, in³ (cm³)

a , in	b , in	t	A	w	I_{1-1}	k_{1-1}	Z_{1-1}	I_{2-2}	k_{2-2}	Z_{2-2}	x
3	1.410	0.170	1.21	4.1	1.66	1.17	1.10	0.197	0.404	0.202	0.436
3	1.498	0.258	1.47	5.0	1.85	1.12	1.24	0.247	0.410	0.233	0.438
3	1.596	0.356	1.76	6.0	2.07	1.08	1.38	0.305	0.416	0.268	0.455
4	1.580	0.180	1.57	5.4	3.85	1.56	1.93	0.319	0.449	0.283	0.457
4	1.720	0.321	2.13	7.25	4.59	1.47	2.29	0.433	0.450	0.343	0.459
5	1.750	0.190	1.97	6.7	7.49	1.95	3.00	0.479	0.493	0.378	0.484
5	1.885	0.325	2.64	9.0	8.90	1.83	3.56	0.632	0.489	0.450	0.478
6	1.920	0.200	2.40	8.2	13.1	2.34	4.38	0.693	0.537	0.492	0.511
6	2.034	0.314	3.09	10.5	15.2	2.22	5.06	0.866	0.529	0.564	0.499
6	2.157	0.437	3.83	13.0	17.4	2.13	5.80	1.05	0.525	0.642	0.514
7	2.090	0.210	2.87	9.8	21.3	2.72	6.08	0.968	0.581	0.625	0.540
7	2.194	0.314	3.60	12.25	24.2	2.60	6.93	1.17	0.571	0.703	0.525
7	2.299	0.419	4.33	14.75	27.2	2.51	7.78	1.38	0.564	0.779	0.532
8	2.260	0.220	3.36	11.5	32.3	3.10	8.10	1.30	0.625	0.781	0.571
8	2.343	0.303	4.04	13.75	36.2	2.99	9.03	1.53	0.615	0.854	0.553
8	2.527	0.487	5.51	18.75	44.0	2.82	11.0	1.98	0.599	1.01	0.565
9	2.430	0.230	3.91	13.4	47.7	3.49	10.6	1.75	0.669	0.962	0.601
9	2.485	0.285	4.41	15.0	51.0	3.40	11.3	1.93	0.661	1.01	0.586
9	2.648	0.448	5.88	20.0	60.9	3.22	13.5	2.42	0.647	1.17	0.583
10	2.600	0.240	4.49	15.3	67.4	3.87	13.5	2.28	0.713	1.16	0.634
10	2.739	0.379	5.88	20.0	78.9	3.66	15.8	2.81	0.693	1.32	0.606
10	2.886	0.526	7.35	25.0	91.2	3.52	18.2	3.36	0.676	1.48	0.617
10	3.033	0.673	8.82	30.0	103	3.43	20.7	3.95	0.669	1.66	0.649
12	3.047	0.387	7.35	25.0	144	4.43	24.1	4.47	0.780	1.89	0.674
12	3.170	0.510	8.82	30.0	162	4.29	27.0	5.14	0.763	2.06	0.674

Table A-7Properties of Structural-Steel Channels (*Continued*)

$a \times b, \text{ mm}$	m	t	A	I_{1-1}	k_{1-1}	Z_{1-1}	I_{2-2}	k_{2-2}	Z_{2-2}	x
76 × 38	6.70	5.1	8.53	74.14	2.95	19.46	10.66	1.12	4.07	1.19
102 × 51	10.42	6.1	13.28	207.7	3.95	40.89	29.10	1.48	8.16	1.51
127 × 64	14.90	6.4	18.98	482.5	5.04	75.99	67.23	1.88	15.25	1.94
152 × 76	17.88	6.4	22.77	851.5	6.12	111.8	113.8	2.24	21.05	2.21
152 × 89	23.84	7.1	30.36	1166	6.20	153.0	215.1	2.66	35.70	2.86
178 × 76	20.84	6.6	26.54	1337	7.10	150.4	134.0	2.25	24.72	2.20
178 × 89	26.81	7.6	34.15	1753	7.16	197.2	241.0	2.66	39.29	2.76
203 × 76	23.82	7.1	30.34	1950	8.02	192.0	151.3	2.23	27.59	2.13
203 × 89	29.78	8.1	37.94	2491	8.10	245.2	264.4	2.64	42.34	2.65
229 × 76	26.06	7.6	33.20	2610	8.87	228.3	158.7	2.19	28.22	2.00
229 × 89	32.76	8.6	41.73	3387	9.01	296.4	285.0	2.61	44.82	2.53
254 × 76	28.29	8.1	36.03	3367	9.67	265.1	162.6	2.12	28.21	1.86
254 × 89	35.74	9.1	45.42	4448	9.88	350.2	302.4	2.58	46.70	2.42
305 × 89	41.69	10.2	53.11	7061	11.5	463.3	325.4	2.48	48.49	2.18
305 × 102	46.18	10.2	58.83	8214	11.8	539.0	499.5	2.91	66.59	2.66

*These sizes are also available in aluminum alloy.

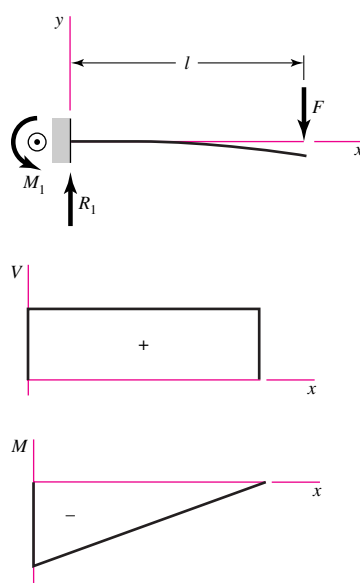
Table A-8Properties of Round
Tubing w_a = unit weight of aluminum tubing, lbf/ft w_s = unit weight of steel tubing, lbf/ft m = unit mass, kg/m A = area, in² (cm²) I = second moment of area, in⁴ (cm⁴) J = second polar moment of area, in⁴ (cm⁴) k = radius of gyration, in (cm) Z = section modulus, in³ (cm³) d, t = size (OD) and thickness, in (mm)

Size, in	w_a	w_s	A	I	k	Z	J
$1 \times \frac{1}{8}$	0.416	1.128	0.344	0.034	0.313	0.067	0.067
$1 \times \frac{1}{4}$	0.713	2.003	0.589	0.046	0.280	0.092	0.092
$1\frac{1}{2} \times \frac{1}{8}$	0.653	1.769	0.540	0.129	0.488	0.172	0.257
$1\frac{1}{2} \times \frac{1}{4}$	1.188	3.338	0.982	0.199	0.451	0.266	0.399
$2 \times \frac{1}{8}$	0.891	2.670	0.736	0.325	0.664	0.325	0.650
$2 \times \frac{1}{4}$	1.663	4.673	1.374	0.537	0.625	0.537	1.074
$2\frac{1}{2} \times \frac{1}{8}$	1.129	3.050	0.933	0.660	0.841	0.528	1.319
$2\frac{1}{2} \times \frac{1}{4}$	2.138	6.008	1.767	1.132	0.800	0.906	2.276
$3 \times \frac{1}{4}$	2.614	7.343	2.160	2.059	0.976	1.373	4.117
$3 \times \frac{3}{8}$	3.742	10.51	3.093	2.718	0.938	1.812	5.436
$4 \times \frac{3}{16}$	2.717	7.654	2.246	4.090	1.350	2.045	8.180
$4 \times \frac{3}{8}$	5.167	14.52	4.271	7.090	1.289	3.544	14.180

Size, mm	m	A	I	k	Z	J
12×2	0.490	0.628	0.082	0.361	0.136	0.163
16×2	0.687	0.879	0.220	0.500	0.275	0.440
16×3	0.956	1.225	0.273	0.472	0.341	0.545
20×4	1.569	2.010	0.684	0.583	0.684	1.367
25×4	2.060	2.638	1.508	0.756	1.206	3.015
25×5	2.452	3.140	1.669	0.729	1.336	3.338
30×4	2.550	3.266	2.827	0.930	1.885	5.652
30×5	3.065	3.925	3.192	0.901	2.128	6.381
42×4	3.727	4.773	8.717	1.351	4.151	17.430
42×5	4.536	5.809	10.130	1.320	4.825	20.255
50×4	4.512	5.778	15.409	1.632	6.164	30.810
50×5	5.517	7.065	18.118	1.601	7.247	36.226

Table A-9

Shear, Moment, and Deflection of Beams
(Note: Force and moment reactions are positive in the directions shown; equations for shear force V and bending moment M follow the sign conventions given in Sec. 4-2.)

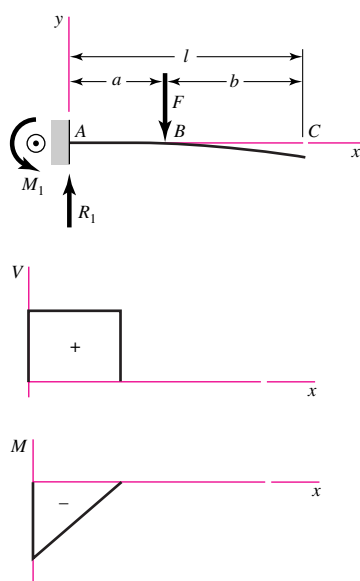
1 Cantilever—end load

$$R_1 = V = F \quad M_1 = Fl$$

$$M = F(x - l)$$

$$y = \frac{Fx^2}{6EI}(x - 3l)$$

$$y_{\max} = -\frac{Fl^3}{3EI}$$

2 Cantilever—intermediate load

$$R_1 = V = F \quad M_1 = Fa$$

$$M_{AB} = F(x - a) \quad M_{BC} = 0$$

$$y_{AB} = \frac{Fx^2}{6EI}(x - 3a)$$

$$y_{BC} = \frac{Fa^2}{6EI}(a - 3x)$$

$$y_{\max} = \frac{Fa^2}{6EI}(a - 3l)$$

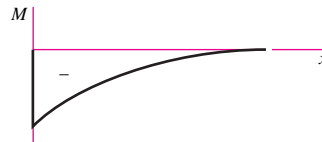
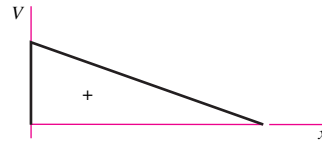
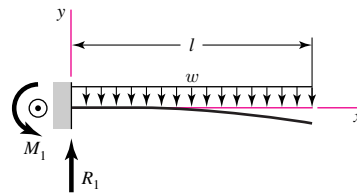
(continued)

Table A-9

Shear, Moment, and Deflection of Beams

(Continued)

(Note: Force and moment reactions are positive in the directions shown; equations for shear force V and bending moment M follow the sign conventions given in Sec. 4-2.)

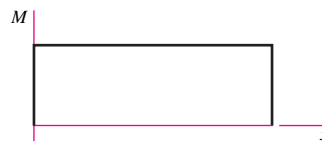
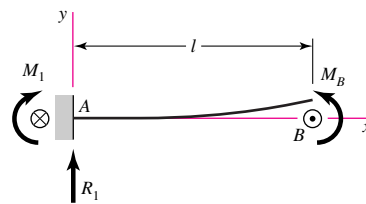
3 Cantilever—uniform load

$$R_1 = wl \quad M_1 = \frac{wl^2}{2}$$

$$V = w(l - x) \quad M = -\frac{w}{2}(l - x)^2$$

$$y = \frac{wx^2}{24EI}(4lx - x^2 - 6l^2)$$

$$y_{\max} = -\frac{wl^4}{8EI}$$

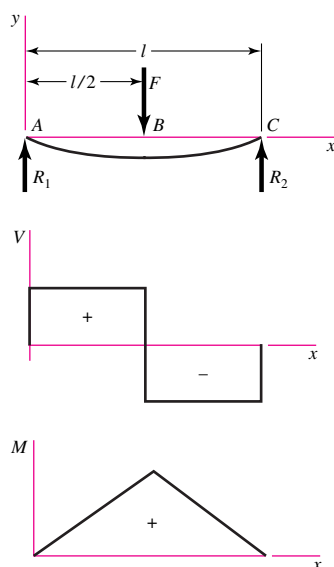
4 Cantilever—moment load

$$R_1 = 0 \quad M_1 = M_B \quad M = M_B$$

$$y = \frac{M_B x^2}{2EI} \quad y_{\max} = \frac{M_B l^2}{2EI}$$

Table A-9

Shear, Moment, and Deflection of Beams
(Continued)
(Note: Force and moment reactions are positive in the directions shown; equations for shear force V and bending moment M follow the sign conventions given in Sec. 4-2.)

5 Simple supports—center load

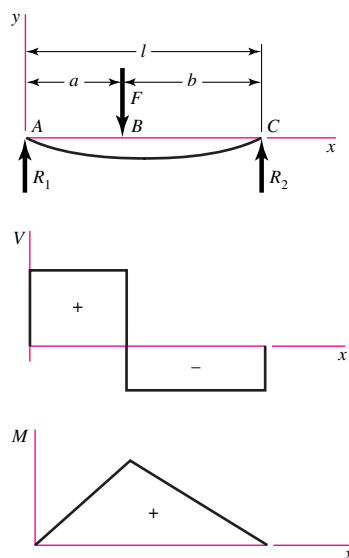
$$R_1 = R_2 = \frac{F}{2} \quad V_{AB} = R_1$$

$$V_{AB} = R_1 \quad V_{BC} = -R_2$$

$$M_{AB} = \frac{Fx}{2} \quad M_{BC} = \frac{F}{2}(l - x)$$

$$y_{AB} = \frac{Fx}{48EI}(4x^2 - 3l^2)$$

$$y_{\max} = -\frac{Fl^3}{48EI}$$

6 Simple supports—intermediate load

$$R_1 = \frac{Fb}{l} \quad R_2 = \frac{Fa}{l}$$

$$V_{AB} = R_1 \quad V_{BC} = -R_2$$

$$M_{AB} = \frac{Fbx}{l} \quad M_{BC} = \frac{Fa}{l}(l - x)$$

$$y_{AB} = \frac{Fbx}{6EI}(x^2 + b^2 - l^2)$$

$$y_{BC} = \frac{Fa(l - x)}{6EI}(x^2 + a^2 - 2lx)$$

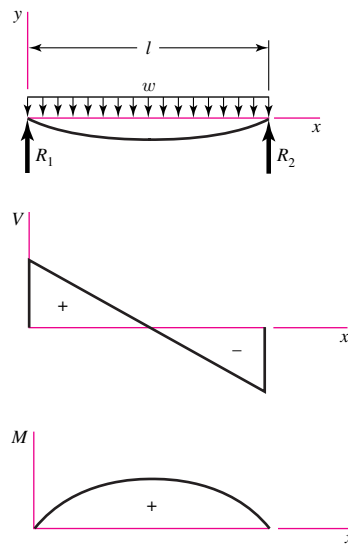
(continued)

Table A-9

Shear, Moment, and Deflection of Beams

(Continued)

(Note: Force and moment reactions are positive in the directions shown; equations for shear force V and bending moment M follow the sign conventions given in Sec. 4-2.)

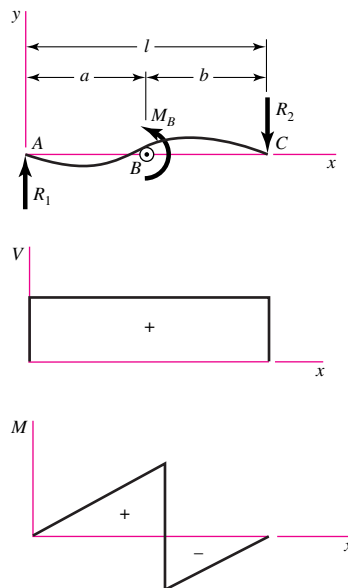
7 Simple supports—uniform load

$$R_1 = R_2 = \frac{wl}{2} \quad V = \frac{wl}{2} - wx$$

$$M = \frac{wx}{2}(l - x)$$

$$y = \frac{wx}{24EI}(2lx^2 - x^3 - l^3)$$

$$y_{\max} = -\frac{5wl^4}{384EI}$$

8 Simple supports—moment load

$$R_1 = R_2 = \frac{M_B}{l} \quad V = \frac{M_B}{l}$$

$$M_{AB} = \frac{M_B x}{l} \quad M_{BC} = \frac{M_B}{l}(x - l)$$

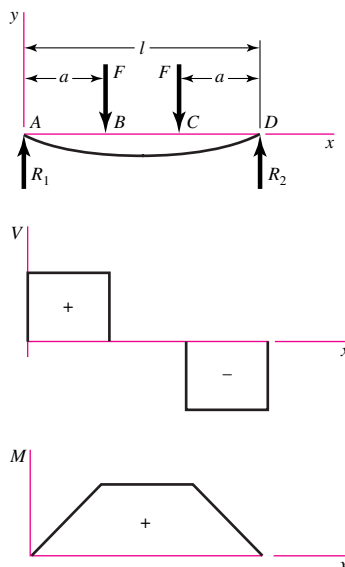
$$y_{AB} = \frac{M_B x}{6EI l}(x^2 + 3a^2 - 6al + 2l^2)$$

$$y_{BC} = \frac{M_B}{6EI l}[x^3 - 3lx^2 + x(2l^2 + 3a^2) - 3a^2l]$$

Table A-9

Shear, Moment, and
Deflection of Beams
(Continued)

(Note: Force and
moment reactions are
positive in the directions
shown; equations for
shear force V and
bending moment M
follow the sign
conventions given in
Sec. 4-2.)

9 Simple supports—twin loads

$$R_1 = R_2 = F \quad V_{AB} = F \quad V_{BC} = 0$$

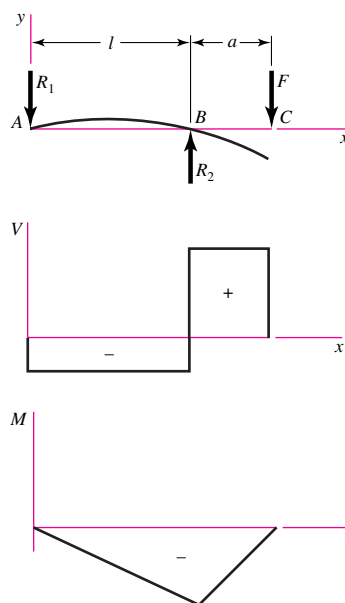
$$V_{CD} = -F$$

$$M_{AB} = Fx \quad M_{BC} = Fa \quad M_{CD} = F(l - x)$$

$$y_{AB} = \frac{Fx}{6EI}(x^2 + 3a^2 - 3la)$$

$$y_{BC} = \frac{Fa}{6EI}(3x^2 + a^2 - 3lx)$$

$$y_{\max} = \frac{Fa}{24EI}(4a^2 - 3l^2)$$

10 Simple supports—overhanging load

$$R_1 = \frac{Fa}{l} \quad R_2 = \frac{F}{l}(l + a)$$

$$V_{AB} = -\frac{Fa}{l} \quad V_{BC} = F$$

$$M_{AB} = -\frac{Fax}{l} \quad M_{BC} = F(x - l - a)$$

$$y_{AB} = \frac{Fax}{6EI}(l^2 - x^2)$$

$$y_{BC} = \frac{F(x - l)}{6EI}[(x - l)^2 - a(3x - l)]$$

$$y_c = -\frac{Fa^2}{3EI}(l + a)$$

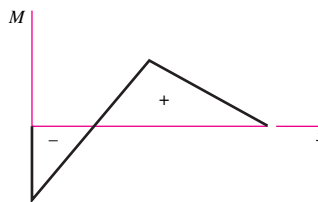
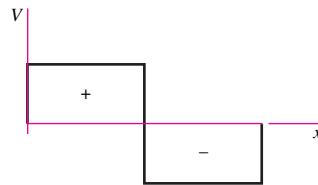
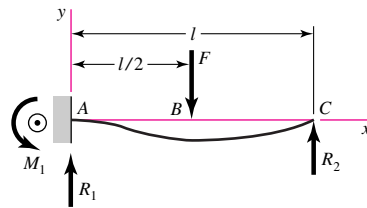
(continued)

Table A-9

Shear, Moment, and
Deflection of Beams

(Continued)

(Note: Force and moment reactions are positive in the directions shown; equations for shear force V and bending moment M follow the sign conventions given in Sec. 4-2.)

11 One fixed and one simple support—center load

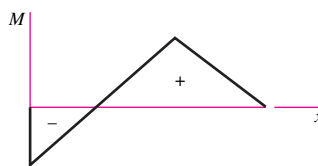
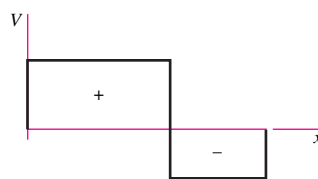
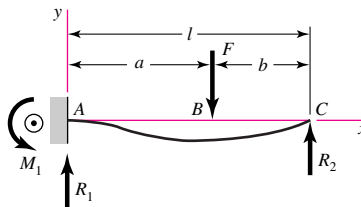
$$R_1 = \frac{11F}{16} \quad R_2 = \frac{5F}{16} \quad M_1 = \frac{3Fl}{16}$$

$$V_{AB} = R_1 \quad V_{BC} = -R_2$$

$$M_{AB} = \frac{F}{16}(11x - 3l) \quad M_{BC} = \frac{5F}{16}(l - x)$$

$$y_{AB} = \frac{Fx^2}{96EI}(11x - 9l)$$

$$y_{BC} = \frac{F(l-x)}{96EI}(5x^2 + 2l^2 - 10lx)$$

12 One fixed and one simple support—intermediate load

$$R_1 = \frac{Fb}{2l^3}(3l^2 - b^2) \quad R_2 = \frac{Fa^2}{2l^3}(3l - a)$$

$$M_1 = \frac{Fb}{2l^2}(l^2 - b^2)$$

$$V_{AB} = R_1 \quad V_{BC} = -R_2$$

$$M_{AB} = \frac{Fb}{2l^3}[b^2l - l^3 + x(3l^2 - b^2)]$$

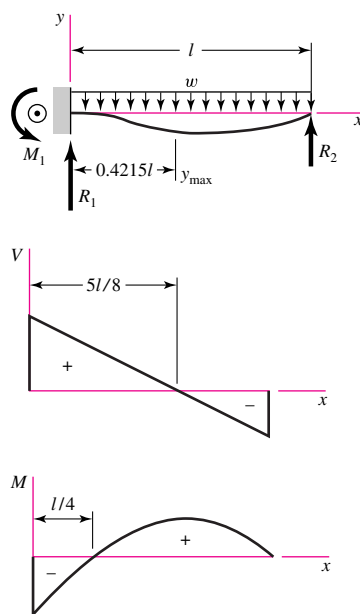
$$M_{BC} = \frac{Fa^2}{2l^3}(3l^2 - 3lx - al + ax)$$

$$y_{AB} = \frac{Fbx^2}{12EI l^3}[3l(b^2 - l^2) + x(3l^2 - b^2)]$$

$$y_{BC} = y_{AB} - \frac{F(x-a)^3}{6EI}$$

Table A-9Shear, Moment, and
Deflection of Beams*(Continued)*

(Note: Force and moment reactions are positive in the directions shown; equations for shear force V and bending moment M follow the sign conventions given in Sec. 4-2.)

13 One fixed and one simple support—uniform load

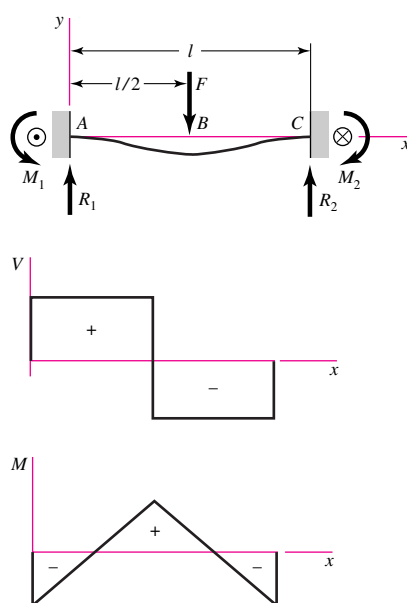
$$R_1 = \frac{5wl}{8} \quad R_2 = \frac{3wl}{8} \quad M_1 = \frac{wl^2}{8}$$

$$V = \frac{5wl}{8} - wx$$

$$M = -\frac{w}{8}(4x^2 - 5lx + l^2)$$

$$y = \frac{wx^2}{48EI}(l-x)(2x-3l)$$

$$y_{\max} = -\frac{wl^4}{185EI}$$

14 Fixed supports—center load

$$R_1 = R_2 = \frac{F}{2} \quad M_1 = M_2 = \frac{Fl}{8}$$

$$V_{AB} = -V_{BC} = \frac{F}{2}$$

$$M_{AB} = \frac{F}{8}(4x - l) \quad M_{BC} = \frac{F}{8}(3l - 4x)$$

$$y_{AB} = \frac{Fx^2}{48EI}(4x - 3l)$$

$$y_{\max} = -\frac{Fl^3}{192EI}$$

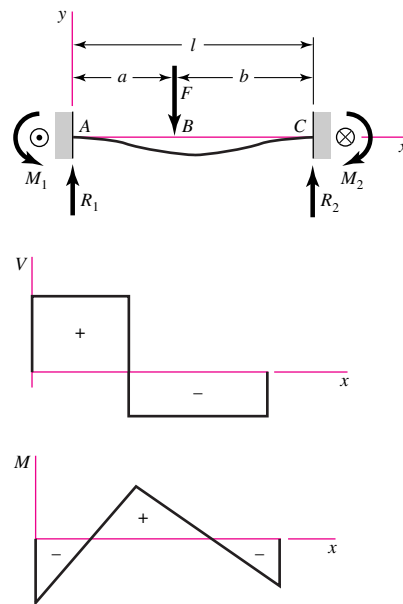
(continued)

Table A-9

Shear, Moment, and Deflection of Beams

(Continued)

(Note: Force and moment reactions are positive in the directions shown; equations for shear force V and bending moment M follow the sign conventions given in Sec. 4-2.)

15 Fixed supports—intermediate load

$$R_1 = \frac{Fb^2}{l^3}(3a + b) \quad R_2 = \frac{Fa^2}{l^3}(3b + a)$$

$$M_1 = \frac{Fab^2}{l^2} \quad M_2 = \frac{Fa^2b}{l^2}$$

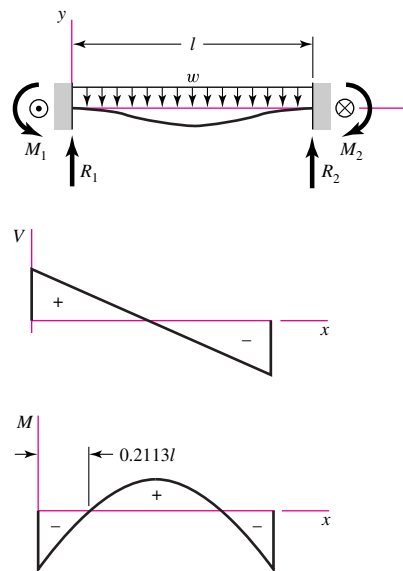
$$V_{AB} = R_1 \quad V_{BC} = -R_2$$

$$M_{AB} = \frac{Fb^2}{l^3}[x(3a + b) - al]$$

$$M_{BC} = M_{AB} - F(x - a)$$

$$y_{AB} = \frac{Fb^2x^2}{6EI^3}[x(3a + b) - 3al]$$

$$y_{BC} = \frac{Fa^2(l - x)^2}{6EI^3}[(l - x)(3b + a) - 3bl]$$

16 Fixed supports—uniform load

$$R_1 = R_2 = \frac{wl}{2} \quad M_1 = M_2 = \frac{wl^2}{12}$$

$$V = \frac{w}{2}(l - 2x)$$

$$M = \frac{w}{12}(6lx - 6x^2 - l^2)$$

$$y = -\frac{wx^2}{24EI}(l - x)^2$$

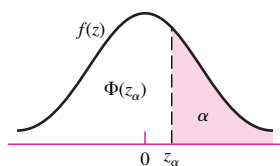
$$y_{\max} = -\frac{wl^4}{384EI}$$

Table A-10

Cumulative Distribution Function of Normal (Gaussian) Distribution

$$\Phi(z_\alpha) = \int_{-\infty}^{z_\alpha} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) du$$

$$= \begin{cases} \alpha & z_\alpha \leq 0 \\ 1 - \alpha & z_\alpha > 0 \end{cases}$$



z_α	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3238	0.3192	0.3156	0.3121
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
2.3	0.0107	0.0104	0.0102	0.00990	0.00964	0.00939	0.00914	0.00889	0.00866	0.00842
2.4	0.00820	0.00798	0.00776	0.00755	0.00734	0.00714	0.00695	0.00676	0.00657	0.00639
2.5	0.00621	0.00604	0.00587	0.00570	0.00554	0.00539	0.00523	0.00508	0.00494	0.00480
2.6	0.00466	0.00453	0.00440	0.00427	0.00415	0.00402	0.00391	0.00379	0.00368	0.00357
2.7	0.00347	0.00336	0.00326	0.00317	0.00307	0.00298	0.00289	0.00280	0.00272	0.00264
2.8	0.00256	0.00248	0.00240	0.00233	0.00226	0.00219	0.00212	0.00205	0.00199	0.00193
2.9	0.00187	0.00181	0.00175	0.00169	0.00164	0.00159	0.00154	0.00149	0.00144	0.00139

(continued)

Table A-10Cumulative Distribution Function of Normal (Gaussian) Distribution (*Continued*)

Z_α	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
3	0.00135	0.0 ³ 968	0.0 ³ 687	0.0 ³ 483	0.0 ³ 337	0.0 ³ 233	0.0 ³ 159	0.0 ³ 108	0.0 ⁴ 723	0.0 ⁴ 481
4	0.0 ⁴ 317	0.0 ⁴ 207	0.0 ⁴ 133	0.0 ⁵ 854	0.0 ⁵ 541	0.0 ⁵ 340	0.0 ⁵ 211	0.0 ⁵ 130	0.0 ⁶ 793	0.0 ⁶ 479
5	0.0 ⁶ 287	0.0 ⁶ 170	0.0 ⁷ 996	0.0 ⁷ 579	0.0 ⁷ 333	0.0 ⁷ 190	0.0 ⁷ 107	0.0 ⁸ 599	0.0 ⁸ 332	0.0 ⁸ 182
6	0.0 ⁹ 987	0.0 ⁹ 530	0.0 ⁹ 282	0.0 ⁹ 149	0.0 ¹⁰ 777	0.0 ¹⁰ 402	0.0 ¹⁰ 206	0.0 ¹⁰ 104	0.0 ¹¹ 523	0.0 ¹¹ 260
z_α	-1.282	-1.643	-1.960	-2.326	-2.576	-3.090	-3.291	-3.891	-4.417	
$F(z_\alpha)$	0.10	0.05	0.025	0.010	0.005	0.001	0.005	0.00005	0.000005	
$R(z_\alpha)$	0.90	0.95	0.975	0.999	0.995	0.999	0.9995	0.9999	0.999995	

Table A-11

A Selection of
International Tolerance
Grades—Metric Series
(Size Ranges Are for
Over the Lower Limit
and Including the Upper
Limit. All Values Are
in Millimeters)

Source: *Preferred Metric Limits
and Fits*, ANSI B4.2-1978.
See also BSI 4500.

Basic Sizes	Tolerance Grades					
	IT6	IT7	IT8	IT9	IT10	IT11
0-3	0.006	0.010	0.014	0.025	0.040	0.060
3-6	0.008	0.012	0.018	0.030	0.048	0.075
6-10	0.009	0.015	0.022	0.036	0.058	0.090
10-18	0.011	0.018	0.027	0.043	0.070	0.110
18-30	0.013	0.021	0.033	0.052	0.084	0.130
30-50	0.016	0.025	0.039	0.062	0.100	0.160
50-80	0.019	0.030	0.046	0.074	0.120	0.190
80-120	0.022	0.035	0.054	0.087	0.140	0.220
120-180	0.025	0.040	0.063	0.100	0.160	0.250
180-250	0.029	0.046	0.072	0.115	0.185	0.290
250-315	0.032	0.052	0.081	0.130	0.210	0.320
315-400	0.036	0.057	0.089	0.140	0.230	0.360

Table A-12

Fundamental Deviations for Shafts—Metric Series

(Size Ranges Are for *Over* the Lower Limit and *Including* the Upper Limit. All Values Are in Millimeters)*Source: Preferred Metric Limits and Fits*, ANSI B4.2-1978. See also BSI 4500.

Basic Sizes	Upper-Deviation Letter					Lower-Deviation Letter				
	c	d	f	g	h	k	n	p	s	u
0–3	–0.060	–0.020	–0.006	–0.002	0	0	+0.004	+0.006	+0.014	+0.018
3–6	–0.070	–0.030	–0.010	–0.004	0	+0.001	+0.008	+0.012	+0.019	+0.023
6–10	–0.080	–0.040	–0.013	–0.005	0	+0.001	+0.010	+0.015	+0.023	+0.028
10–14	–0.095	–0.050	–0.016	–0.006	0	+0.001	+0.012	+0.018	+0.028	+0.033
14–18	–0.095	–0.050	–0.016	–0.006	0	+0.001	+0.012	+0.018	+0.028	+0.033
18–24	–0.110	–0.065	–0.020	–0.007	0	+0.002	+0.015	+0.022	+0.035	+0.041
24–30	–0.110	–0.065	–0.020	–0.007	0	+0.002	+0.015	+0.022	+0.035	+0.048
30–40	–0.120	–0.080	–0.025	–0.009	0	+0.002	+0.017	+0.026	+0.043	+0.060
40–50	–0.130	–0.080	–0.025	–0.009	0	+0.002	+0.017	+0.026	+0.043	+0.070
50–65	–0.140	–0.100	–0.030	–0.010	0	+0.002	+0.020	+0.032	+0.053	+0.087
65–80	–0.150	–0.100	–0.030	–0.010	0	+0.002	+0.020	+0.032	+0.059	+0.102
80–100	–0.170	–0.120	–0.036	–0.012	0	+0.003	+0.023	+0.037	+0.071	+0.124
100–120	–0.180	–0.120	–0.036	–0.012	0	+0.003	+0.023	+0.037	+0.079	+0.144
120–140	–0.200	–0.145	–0.043	–0.014	0	+0.003	+0.027	+0.043	+0.092	+0.170
140–160	–0.210	–0.145	–0.043	–0.014	0	+0.003	+0.027	+0.043	+0.100	+0.190
160–180	–0.230	–0.145	–0.043	–0.014	0	+0.003	+0.027	+0.043	+0.108	+0.210
180–200	–0.240	–0.170	–0.050	–0.015	0	+0.004	+0.031	+0.050	+0.122	+0.236
200–225	–0.260	–0.170	–0.050	–0.015	0	+0.004	+0.031	+0.050	+0.130	+0.258
225–250	–0.280	–0.170	–0.050	–0.015	0	+0.004	+0.031	+0.050	+0.140	+0.284
250–280	–0.300	–0.190	–0.056	–0.017	0	+0.004	+0.034	+0.056	+0.158	+0.315
280–315	–0.330	–0.190	–0.056	–0.017	0	+0.004	+0.034	+0.056	+0.170	+0.350
315–355	–0.360	–0.210	–0.062	–0.018	0	+0.004	+0.037	+0.062	+0.190	+0.390
355–400	–0.400	–0.210	–0.062	–0.018	0	+0.004	+0.037	+0.062	+0.208	+0.435

Table A-13

A Selection of
International Tolerance
Grades—Inch Series
(Size Ranges Are for
Over the Lower Limit
and *Including* the Upper
Limit. All Values Are in
Inches, Converted from
Table A-11)

Basic Sizes	Tolerance Grades					
	IT6	IT7	IT8	IT9	IT10	IT11
0–0.12	0.0002	0.0004	0.0006	0.0010	0.0016	0.0024
0.12–0.24	0.0003	0.0005	0.0007	0.0012	0.0019	0.0030
0.24–0.40	0.0004	0.0006	0.0009	0.0014	0.0023	0.0035
0.40–0.72	0.0004	0.0007	0.0011	0.0017	0.0028	0.0043
0.72–1.20	0.0005	0.0008	0.0013	0.0020	0.0033	0.0051
1.20–2.00	0.0006	0.0010	0.0015	0.0024	0.0039	0.0063
2.00–3.20	0.0007	0.0012	0.0018	0.0029	0.0047	0.0075
3.20–4.80	0.0009	0.0014	0.0021	0.0034	0.0055	0.0087
4.80–7.20	0.0010	0.0016	0.0025	0.0039	0.0063	0.0098
7.20–10.00	0.0011	0.0018	0.0028	0.0045	0.0073	0.0114
10.00–12.60	0.0013	0.0020	0.0032	0.0051	0.0083	0.0126
12.60–16.00	0.0014	0.0022	0.0035	0.0055	0.0091	0.0142

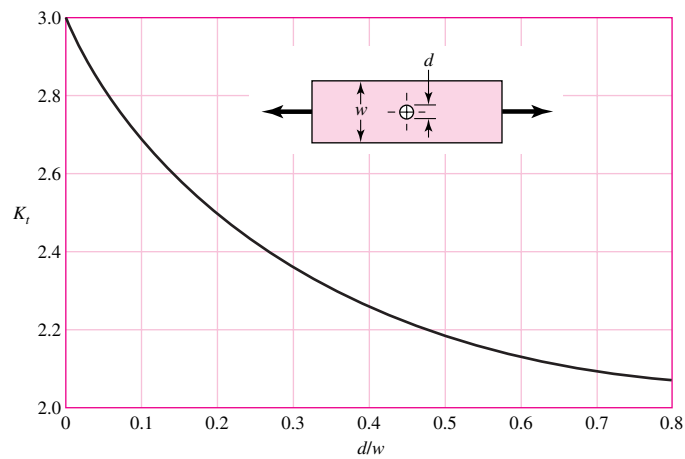
Table A-14

Fundamental Deviations for Shafts—Inch Series (Size Ranges Are for Over the Lower Limit and Including the Upper Limit. All Values Are in Inches, Converted from Table A-12)

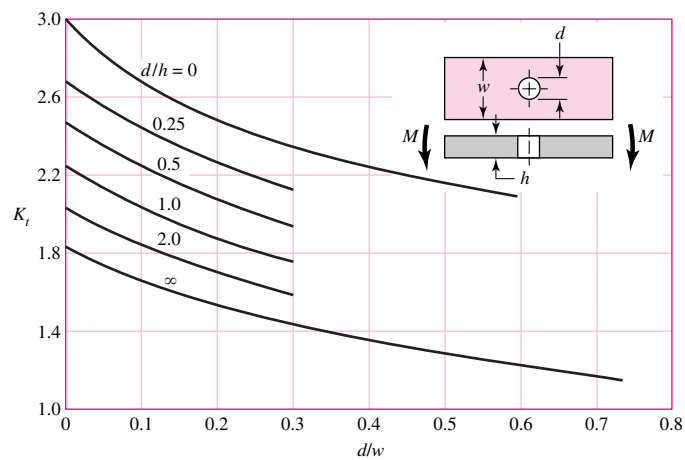
Basic Sizes	Upper-Deviation Letter					Lower-Deviation Letter				
	c	d	f	g	h	k	n	p	s	u
0-0.12	-0.0024	-0.0008	-0.0002	-0.0001	0	0	+0.0002	+0.0002	+0.0006	+0.0007
0.12-0.24	-0.0028	-0.0012	-0.0004	-0.0002	0	0	+0.0003	+0.0005	+0.0007	+0.0009
0.24-0.40	-0.0031	-0.0016	-0.0005	-0.0002	0	0	+0.0004	+0.0006	+0.0009	+0.0011
0.40-0.72	-0.0037	-0.0020	-0.0006	-0.0002	0	0	+0.0005	+0.0007	+0.0011	+0.0013
0.72-0.96	-0.0043	-0.0026	-0.0008	-0.0003	0	+0.0001	+0.0006	+0.0009	+0.0014	+0.0016
0.96-1.20	-0.0043	-0.0026	-0.0008	-0.0003	0	+0.0001	+0.0006	+0.0009	+0.0014	+0.0019
1.20-1.60	-0.0047	-0.0031	-0.0010	-0.0004	0	+0.0001	+0.0007	+0.0010	+0.0017	+0.0024
1.60-2.00	-0.0051	-0.0031	-0.0010	-0.0004	0	+0.0001	+0.0007	+0.0010	+0.0017	+0.0028
2.00-2.60	-0.0055	-0.0039	-0.0012	-0.0004	0	+0.0001	+0.0008	+0.0013	+0.0021	+0.0034
2.60-3.20	-0.0059	-0.0039	-0.0012	-0.0004	0	+0.0001	+0.0008	+0.0013	+0.0023	+0.0040
3.20-4.00	-0.0067	-0.0047	-0.0014	-0.0005	0	+0.0001	+0.0009	+0.0015	+0.0028	+0.0049
4.00-4.80	-0.0071	-0.0047	-0.0014	-0.0005	0	+0.0001	+0.0009	+0.0015	+0.0031	+0.0057
4.80-5.60	-0.0079	-0.0057	-0.0017	-0.0006	0	+0.0001	+0.0011	+0.0017	+0.0036	+0.0067
5.60-6.40	-0.0083	-0.0057	-0.0017	-0.0006	0	+0.0001	+0.0011	+0.0017	+0.0039	+0.0075
6.40-7.20	-0.0091	-0.0057	-0.0017	-0.0006	0	+0.0001	+0.0011	+0.0017	+0.0043	+0.0083
7.20-8.00	-0.0094	-0.0067	-0.0020	-0.0006	0	+0.0002	+0.0012	+0.0020	+0.0048	+0.0093
8.00-9.00	-0.0102	-0.0067	-0.0020	-0.0006	0	+0.0002	+0.0012	+0.0020	+0.0051	+0.0102
9.00-10.00	-0.0110	-0.0067	-0.0020	-0.0006	0	+0.0002	+0.0012	+0.0020	+0.0055	+0.0112
10.00-11.20	-0.0118	-0.0075	-0.0022	-0.0007	0	+0.0002	+0.0013	+0.0022	+0.0062	+0.0124
11.20-12.60	-0.0130	-0.0075	-0.0022	-0.0007	0	+0.0002	+0.0013	+0.0022	+0.0067	+0.0130
12.60-14.20	-0.0142	-0.0083	-0.0024	-0.0007	0	+0.0002	+0.0015	+0.0024	+0.0075	+0.0154
14.20-16.00	-0.0157	-0.0083	-0.0024	-0.0007	0	+0.0002	+0.0015	+0.0024	+0.0082	+0.0171

Table A-15Charts of Theoretical Stress-Concentration Factors K_t^* **Figure A-15-1**

Bar in tension or simple compression with a transverse hole. $\sigma_0 = F/A$, where $A = (w - d)t$ and t is the thickness.

**Figure A-15-2**

Rectangular bar with a transverse hole in bending. $\sigma_0 = Mc/I$, where $I = (w - d)h^3/12$.

**Figure A-15-3**

Notched rectangular bar in tension or simple compression. $\sigma_0 = F/A$, where $A = dt$ and t is the thickness.

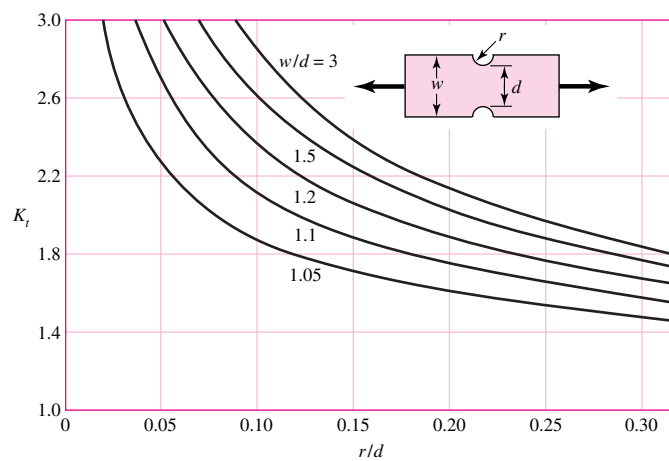
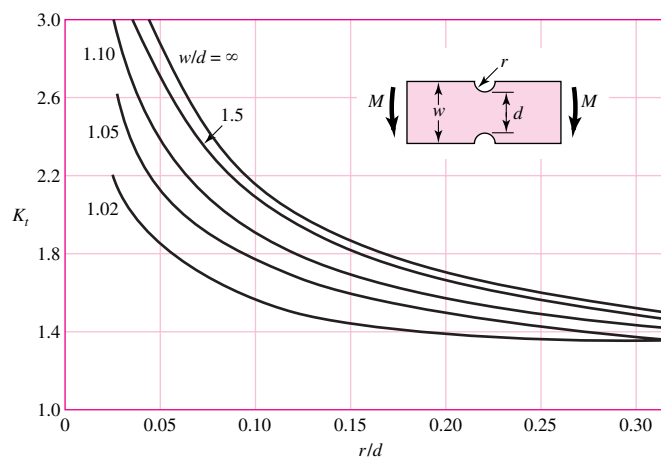
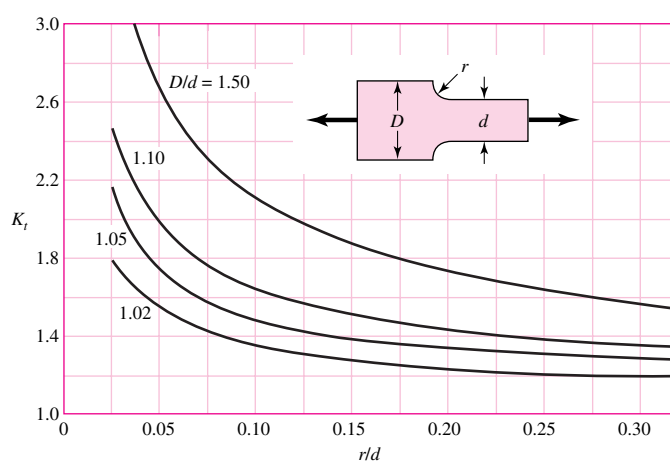


Table A-15Charts of Theoretical Stress-Concentration Factors K_t^* (Continued)**Figure A-15-4**

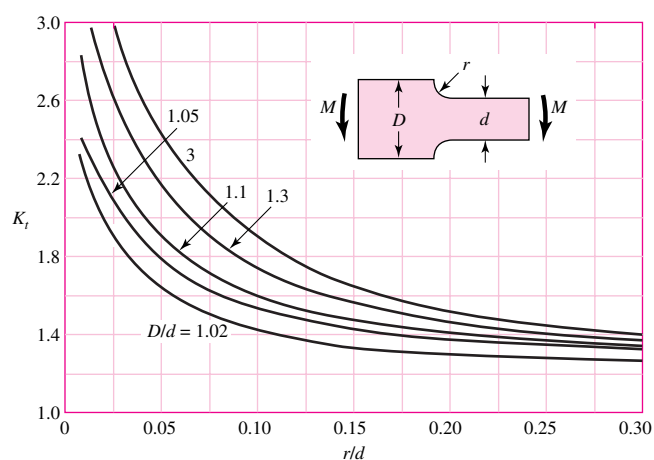
Notched rectangular bar in bending. $\sigma_0 = Mc/I$, where $c = d/2$, $I = td^3/12$, and t is the thickness.

**Figure A-15-5**

Rectangular filleted bar in tension or simple compression. $\sigma_0 = F/A$, where $A = dt$ and t is the thickness.

**Figure A-15-6**

Rectangular filleted bar in bending. $\sigma_0 = Mc/I$, where $c = d/2$, $I = td^3/12$, t is the thickness.

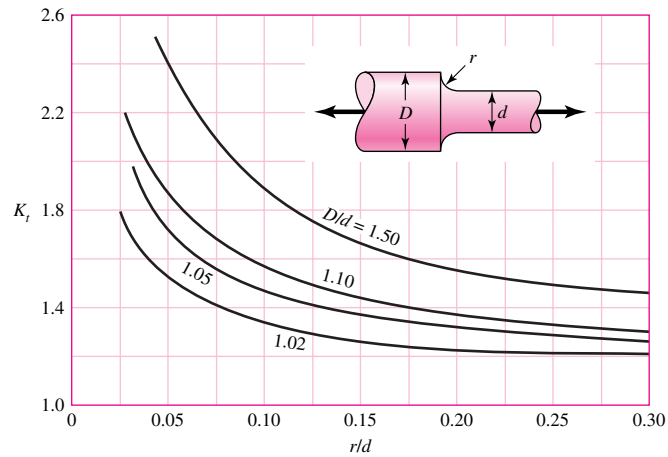


(continued)

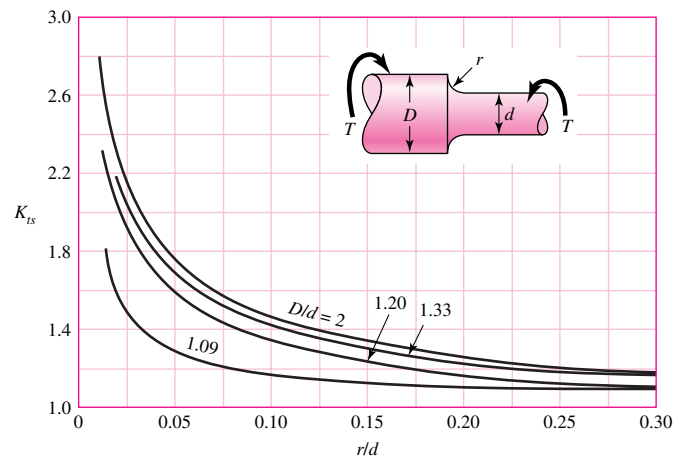
*Factors from R. E. Peterson, "Design Factors for Stress Concentration," *Machine Design*, vol. 23, no. 2, February 1951, p. 169; no. 3, March 1951, p. 161, no. 5, May 1951, p. 159; no. 6, June 1951, p. 173; no. 7, July 1951, p. 155. Reprinted with permission from *Machine Design*, a Penton Media Inc. publication.

Table A-15Charts of Theoretical Stress-Concentration Factors K_t^* (Continued)**Figure A-15-7**

Round shaft with shoulder fillet
in tension. $\sigma_0 = F/A$, where
 $A = \pi d^2/4$.

**Figure A-15-8**

Round shaft with shoulder fillet
in torsion. $\tau_0 = Tc/J$, where
 $c = d/2$ and $J = \pi d^4/32$.

**Figure A-15-9**

Round shaft with shoulder fillet
in bending. $\sigma_0 = Mc/I$, where
 $c = d/2$ and $I = \pi d^4/64$.

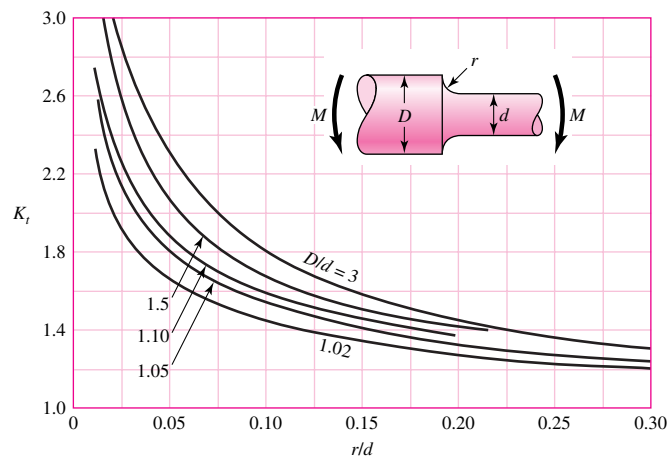
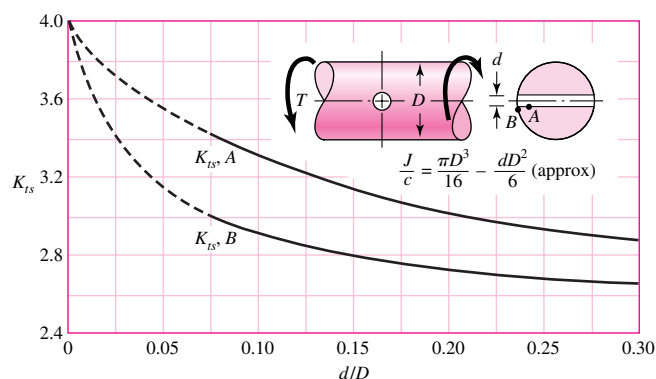


Table A-15Charts of Theoretical Stress-Concentration Factors K_t^* (Continued)**Figure A-15-10**

Round shaft in torsion with transverse hole.

**Figure A-15-11**

Round shaft in bending with a transverse hole. $\sigma_0 = M/[(\pi D^3/32) - (d D^2/6)]$, approximately.

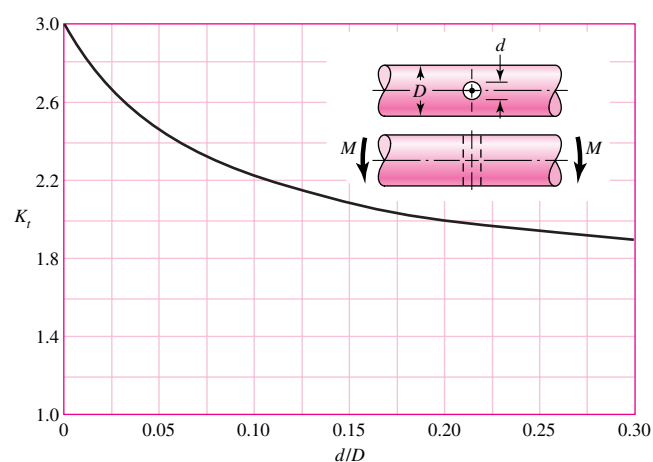
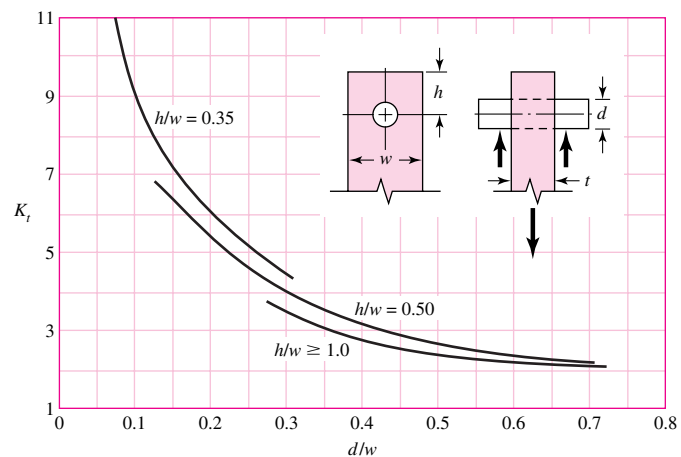
**Figure A-15-12**

Plate loaded in tension by a pin through a hole. $\sigma_0 = F/A$, where $A = (w - d)t$. When clearance exists, increase K_t 35 to 50 percent. (M. M. Frocht and H. N. Hill, "Stress Concentration Factors around a Central Circular Hole in a Plate Loaded through a Pin in Hole," *J. Appl. Mechanics*, vol. 7, no. 1, March 1940, p. A-5.)

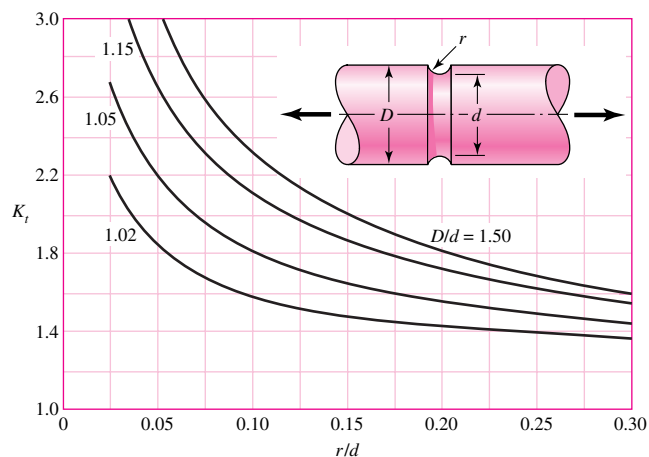


(continued)

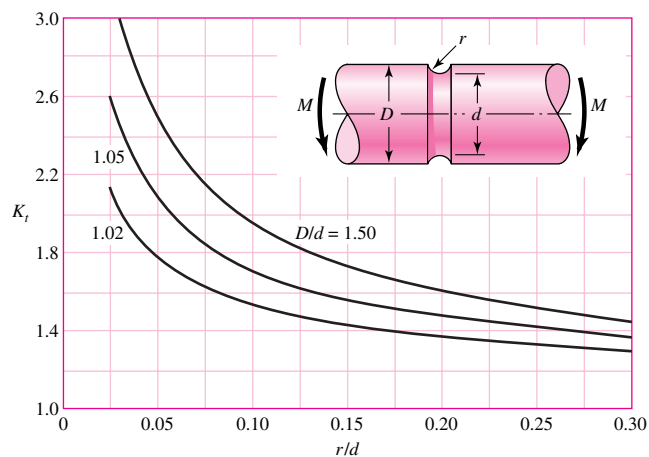
*Factors from R. E. Peterson, "Design Factors for Stress Concentration," *Machine Design*, vol. 23, no. 2, February 1951, p. 169; no. 3, March 1951, p. 161, no. 5, May 1951, p. 159; no. 6, June 1951, p. 173; no. 7, July 1951, p. 155. Reprinted with permission from *Machine Design*, a Penton Media Inc. publication.

Table A-15Charts of Theoretical Stress-Concentration Factors K_t^* (Continued)**Figure A-15-13**

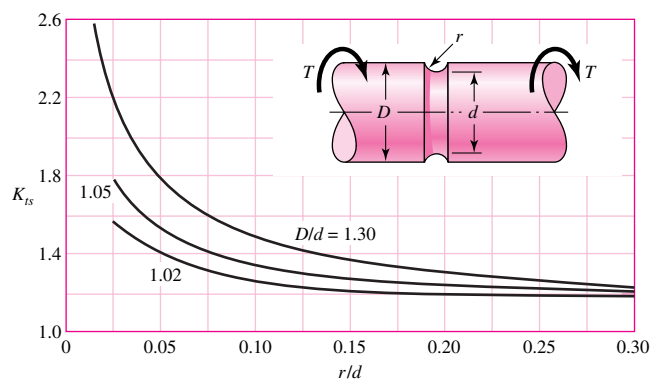
Grooved round bar in tension.
 $\sigma_0 = F/A$, where
 $A = \pi d^2/4$.

**Figure A-15-14**

Grooved round bar in bending. $\sigma_0 = Mc/I$, where
 $c = d/2$ and $I = \pi d^4/64$.

**Figure A-15-15**

Grooved round bar in torsion.
 $\tau_0 = Tc/J$, where $c = d/2$
and $J = \pi d^4/32$.

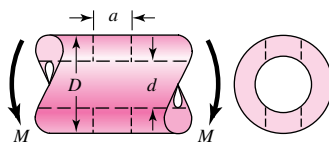


*Factors from R. E. Peterson, "Design Factors for Stress Concentration," *Machine Design*, vol. 23, no. 2, February 1951, p. 169; no. 3, March 1951, p. 161, no. 5, May 1951, p. 159; no. 6, June 1951, p. 173; no. 7, July 1951, p. 155. Reprinted with permission from *Machine Design*, a Penton Media Inc. publication.

Table A-16

Approximate Stress-Concentration Factor K_t for Bending of a Round Bar or Tube with a Transverse Round Hole

Source: R. E. Peterson, *Stress Concentration Factors*, Wiley, New York, 1974, pp. 146, 235.



The nominal bending stress is $\sigma_0 = M/Z_{\text{net}}$ where Z_{net} is a reduced value of the section modulus and is defined by

$$Z_{\text{net}} = \frac{\pi A}{32D}(D^4 - d^4)$$

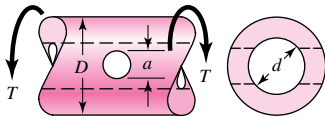
Values of A are listed in the table. Use $d = 0$ for a solid bar

a/D	d/D					
	0.9		0.6		0	
	A	K_t	A	K_t	A	K_t
0.050	0.92	2.63	0.91	2.55	0.88	2.42
0.075	0.89	2.55	0.88	2.43	0.86	2.35
0.10	0.86	2.49	0.85	2.36	0.83	2.27
0.125	0.82	2.41	0.82	2.32	0.80	2.20
0.15	0.79	2.39	0.79	2.29	0.76	2.15
0.175	0.76	2.38	0.75	2.26	0.72	2.10
0.20	0.73	2.39	0.72	2.23	0.68	2.07
0.225	0.69	2.40	0.68	2.21	0.65	2.04
0.25	0.67	2.42	0.64	2.18	0.61	2.00
0.275	0.66	2.48	0.61	2.16	0.58	1.97
0.30	0.64	2.52	0.58	2.14	0.54	1.94

(continued)

Table A-16 (Continued)

Approximate Stress-Concentration Factors K_{ts} for a Round Bar or Tube Having a Transverse Round Hole and Loaded in Torsion Source: R. E. Peterson, *Stress Concentration Factors*, Wiley, New York, 1974, pp. 148, 244.



The maximum stress occurs on the inside of the hole, slightly below the shaft surface. The nominal shear stress is $\tau_0 = TD/2J_{\text{net}}$, where J_{net} is a reduced value of the second polar moment of area and is defined by

$$J_{\text{net}} = \frac{\pi A(D^4 - d^4)}{32}$$

Values of A are listed in the table. Use $d = 0$ for a solid bar.

a/D	d/D									
	0.9		0.8		0.6		0.4		0	
	A	K_{ts}	A	K_{ts}	A	K_{ts}	A	K_{ts}	A	K_{ts}
0.05	0.96	1.78							0.95	1.77
0.075	0.95	1.82							0.93	1.71
0.10	0.94	1.76	0.93	1.74	0.92	1.72	0.92	1.70	0.92	1.68
0.125	0.91	1.76	0.91	1.74	0.90	1.70	0.90	1.67	0.89	1.64
0.15	0.90	1.77	0.89	1.75	0.87	1.69	0.87	1.65	0.87	1.62
0.175	0.89	1.81	0.88	1.76	0.87	1.69	0.86	1.64	0.85	1.60
0.20	0.88	1.96	0.86	1.79	0.85	1.70	0.84	1.63	0.83	1.58
0.25	0.87	2.00	0.82	1.86	0.81	1.72	0.80	1.63	0.79	1.54
0.30	0.80	2.18	0.78	1.97	0.77	1.76	0.75	1.63	0.74	1.51
0.35	0.77	2.41	0.75	2.09	0.72	1.81	0.69	1.63	0.68	1.47
0.40	0.72	2.67	0.71	2.25	0.68	1.89	0.64	1.63	0.63	1.44

Table A-17

Preferred Sizes and
Renard (R-Series)
Numbers
(When a choice can be
made, use one of these
sizes; however, not all
parts or items are
available in all the sizes
shown in the table.)

Fraction of Inches

$\frac{1}{64}, \frac{1}{32}, \frac{1}{16}, \frac{3}{32}, \frac{1}{8}, \frac{5}{32}, \frac{3}{16}, \frac{1}{4}, \frac{5}{16}, \frac{3}{8}, \frac{7}{16}, \frac{1}{2}, \frac{9}{16}, \frac{5}{8}, \frac{11}{16}, \frac{3}{4}, \frac{7}{8}, 1, 1\frac{1}{4}, 1\frac{1}{2}, 1\frac{3}{4}, 2, 2\frac{1}{4},$
 $2\frac{1}{2}, 2\frac{3}{4}, 3, 3\frac{1}{4}, 3\frac{1}{2}, 3\frac{3}{4}, 4, 4\frac{1}{4}, 4\frac{1}{2}, 4\frac{3}{4}, 5, 5\frac{1}{4}, 5\frac{1}{2}, 5\frac{3}{4}, 6, 6\frac{1}{2}, 7, 7\frac{1}{2}, 8, 8\frac{1}{2}, 9, 9\frac{1}{2},$
 $10, 10\frac{1}{2}, 11, 11\frac{1}{2}, 12, 12\frac{1}{2}, 13, 13\frac{1}{2}, 14, 14\frac{1}{2}, 15, 15\frac{1}{2}, 16, 16\frac{1}{2}, 17, 17\frac{1}{2}, 18,$
 $18\frac{1}{2}, 19, 19\frac{1}{2}, 20$

Decimal Inches

0.010, 0.012, 0.016, 0.020, 0.025, 0.032, 0.040, 0.05, 0.06, 0.08, 0.10, 0.12, 0.16,
 0.20, 0.24, 0.30, 0.40, 0.50, 0.60, 0.80, 1.00, 1.20, 1.40, 1.60, 1.80, 2.0, 2.4, 2.6,
 2.8, 3.0, 3.2, 3.4, 3.6, 3.8, 4.0, 4.2, 4.4, 4.6, 4.8, 5.0, 5.2, 5.4, 5.6, 5.8, 6.0, 7.0, 7.5,
 8.5, 9.0, 9.5, 10.0, 10.5, 11.0, 11.5, 12.0, 12.5, 13.0, 13.5, 14.0, 14.5, 15.0, 15.5,
 16.0, 16.5, 17.0, 17.5, 18.0, 18.5, 19.0, 19.5, 20

Millimeters

0.05, 0.06, 0.08, 0.10, 0.12, 0.16, 0.20, 0.25, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80,
 0.90, 1.0, 1.1, 1.2, 1.4, 1.5, 1.6, 1.8, 2.0, 2.2, 2.5, 2.8, 3.0, 3.5, 4.0, 4.5, 5.0, 5.5,
 6.0, 6.5, 7.0, 8.0, 9.0, 10, 11, 12, 14, 16, 18, 20, 22, 25, 28, 30, 32, 35, 40, 45, 50,
 60, 80, 100, 120, 140, 160, 180, 200, 250, 300

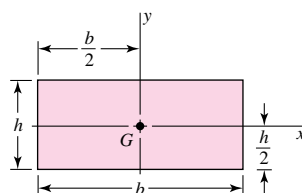
Renard Numbers*

1st choice, R5: 1, 1.6, 2.5, 4, 6.3, 10
 2d choice, R10: 1.25, 2, 3.15, 5, 8
 3d choice, R20: 1.12, 1.4, 1.8, 2.24, 2.8, 3.55, 4.5, 5.6, 7.1, 9
 4th choice, R40: 1.06, 1.18, 1.32, 1.5, 1.7, 1.9, 2.12, 2.36, 2.65, 3, 3.35, 3.75,
 4.25, 4.75, 5.3, 6, 6.7, 7.5, 8.5, 9.5

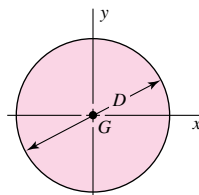
*May be multiplied or divided by powers of 10.

Table A-18

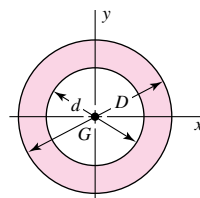
Geometric Properties

Part 1 Properties of Sections A = area G = location of centroid $I_x = \int x^2 dA$ = second moment of area about x axis $I_{xy} = \int xy dA$ = mixed moment of area about x and y axes $J_G = \int r^2 dA = \int (x^2 + y^2) dA = I_x + I_y$
= second polar moment of area about axis through G $k_x^2 = I_x/A$ = squared radius of gyration about x axis**Rectangle**

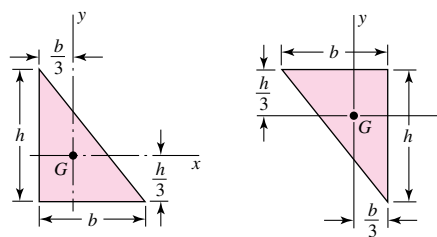
$$A = bh \quad I_x = \frac{bh^3}{12} \quad I_y = \frac{b^3h}{12} \quad I_{xy} = 0$$

Circle

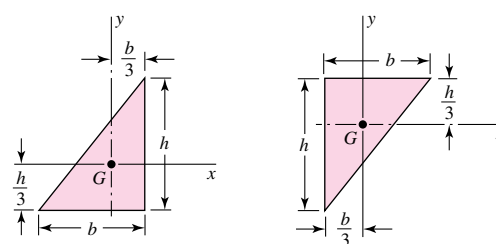
$$A = \frac{\pi D^2}{4} \quad I_x = I_y = \frac{\pi D^4}{64} \quad I_{xy} = 0$$

Hollow circle

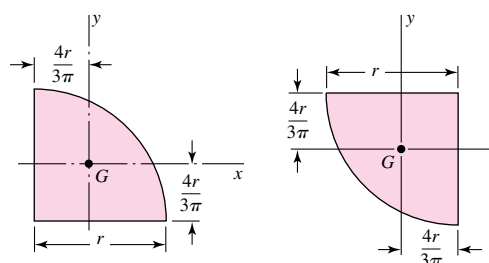
$$A = \frac{\pi}{4}(D^2 - d^2) \quad I_x = I_y = \frac{\pi}{64}(D^4 - d^4) \quad I_{xy} = 0$$

Table A-18Geometric Properties
(Continued)**Right triangles**

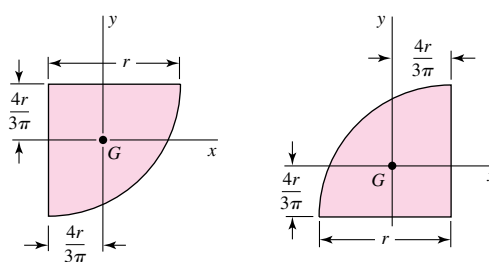
$$A = \frac{bh}{2} \quad I_x = \frac{bh^3}{36} \quad I_y = \frac{b^3h}{36} \quad I_{xy} = \frac{-b^2h^2}{72}$$

Right triangles

$$A = \frac{bh}{2} \quad I_x = \frac{bh^3}{36} \quad I_y = \frac{b^3h}{36} \quad I_{xy} = \frac{b^2h^2}{72}$$

Quarter-circles

$$A = \frac{\pi r^2}{4} \quad I_x = I_y = r^4 \left(\frac{\pi}{16} - \frac{4}{9\pi} \right) \quad I_{xy} = r^4 \left(\frac{1}{8} - \frac{4}{9\pi} \right)$$

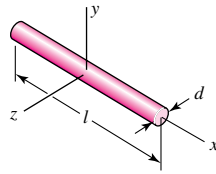
Quarter-circles

$$A = \frac{\pi r^2}{4} \quad I_x = I_y = r^4 \left(\frac{\pi}{16} - \frac{4}{9\pi} \right) \quad I_{xy} = r^4 \left(\frac{4}{9\pi} - \frac{1}{8} \right)$$

(continued)

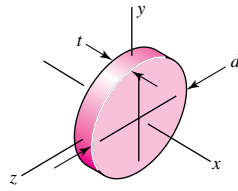
Table A-18Geometric Properties
(Continued)**Part 2 Properties of Solids (ρ = Density, Weight per Unit Volume)**

Rods



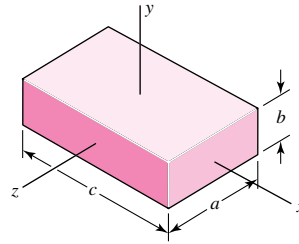
$$m = \frac{\pi d^2 l \rho}{4g} \quad I_y = I_z = \frac{ml^2}{12}$$

Round disks



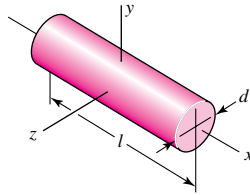
$$m = \frac{\pi d^2 t \rho}{4g} \quad I_x = \frac{md^2}{8} \quad I_y = I_z = \frac{md^2}{16}$$

Rectangular prisms



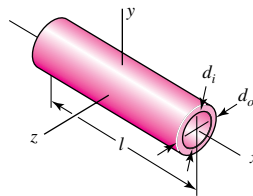
$$m = \frac{abc\rho}{g} \quad I_x = \frac{m}{12}(a^2 + b^2) \quad I_y = \frac{m}{12}(a^2 + c^2) \quad I_z = \frac{m}{12}(b^2 + c^2)$$

Cylinders



$$m = \frac{\pi d^2 l \rho}{4g} \quad I_x = \frac{md^2}{8} \quad I_y = I_z = \frac{m}{48}(3d^2 + 4l^2)$$

Hollow cylinders



$$m = \frac{\pi (d_o^2 - d_i^2) l \rho}{4g} \quad I_x = \frac{m}{8}(d_o^2 + d_i^2) \quad I_y = I_z = \frac{m}{48}(3d_o^2 + 3d_i^2 + 4l^2)$$

Table A-19

American Standard Pipe

Nominal Size, in	Outside Diameter, in	Threads per inch	Wall Thickness, in		
			Standard No. 40	Extra Strong No. 80	Double Extra Strong
$\frac{1}{8}$	0.405	27	0.070	0.098	
$\frac{1}{4}$	0.540	18	0.090	0.122	
$\frac{3}{8}$	0.675	18	0.093	0.129	
$\frac{1}{2}$	0.840	14	0.111	0.151	0.307
$\frac{3}{4}$	1.050	14	0.115	0.157	0.318
1	1.315	11 $\frac{1}{2}$	0.136	0.183	0.369
1 $\frac{1}{4}$	1.660	11 $\frac{1}{2}$	0.143	0.195	0.393
1 $\frac{1}{2}$	1.900	11 $\frac{1}{2}$	0.148	0.204	0.411
2	2.375	11 $\frac{1}{2}$	0.158	0.223	0.447
2 $\frac{1}{2}$	2.875	8	0.208	0.282	0.565
3	3.500	8	0.221	0.306	0.615
3 $\frac{1}{2}$	4.000	8	0.231	0.325	
4	4.500	8	0.242	0.344	0.690
5	5.563	8	0.263	0.383	0.768
6	6.625	8	0.286	0.441	0.884
8	8.625	8	0.329	0.510	0.895

Table A-20

Deterministic ASTM Minimum Tensile and Yield Strengths for Some Hot-Rolled (HR) and Cold-Drawn (CD) Steels [The strengths listed are estimated ASTM minimum values in the size range 18 to 32 mm ($\frac{3}{4}$ to $1\frac{1}{4}$ in). These strengths are suitable for use with the design factor defined in Sec. 1–10, provided the materials conform to ASTM A6 or A568 requirements or are required in the purchase specifications. Remember that a numbering system is not a specification. See Table 1–1 for certain ASTM steels.] *Source:* 1986 SAE Handbook, p. 2.15.

1	2	3	4	5	6	7	8
UNS No.	SAE and/or AISI No.	Process- ing	Tensile Strength, MPa (kpsi)	Yield Strength, MPa (kpsi)	Elongation in 2 in, %	Reduction in Area, %	Brinell Hardness
G10060	1006	HR	300 (43)	170 (24)	30	55	86
		CD	330 (48)	280 (41)	20	45	95
G10100	1010	HR	320 (47)	180 (26)	28	50	95
		CD	370 (53)	300 (44)	20	40	105
G10150	1015	HR	340 (50)	190 (27.5)	28	50	101
		CD	390 (56)	320 (47)	18	40	111
G10180	1018	HR	400 (58)	220 (32)	25	50	116
		CD	440 (64)	370 (54)	15	40	126
G10200	1020	HR	380 (55)	210 (30)	25	50	111
		CD	470 (68)	390 (57)	15	40	131
G10300	1030	HR	470 (68)	260 (37.5)	20	42	137
		CD	520 (76)	440 (64)	12	35	149
G10350	1035	HR	500 (72)	270 (39.5)	18	40	143
		CD	550 (80)	460 (67)	12	35	163
G10400	1040	HR	520 (76)	290 (42)	18	40	149
		CD	590 (85)	490 (71)	12	35	170
G10450	1045	HR	570 (82)	310 (45)	16	40	163
		CD	630 (91)	530 (77)	12	35	179
G10500	1050	HR	620 (90)	340 (49.5)	15	35	179
		CD	690 (100)	580 (84)	10	30	197
G10600	1060	HR	680 (98)	370 (54)	12	30	201
G10800	1080	HR	770 (112)	420 (61.5)	10	25	229
G10950	1095	HR	830 (120)	460 (66)	10	25	248

Table A-21

Mean Mechanical Properties of Some Heat-Treated Steels

[These are typical properties for materials normalized and annealed. The properties for quenched and tempered (Q&T) steels are from a single heat. Because of the many variables, the properties listed are global averages. In all cases, data were obtained from specimens of diameter 0.505 in, machined from 1-in rounds, and of gauge length 2 in. unless noted, all specimens were oil-quenched.] Source: *ASM Metals Reference Book*, 2d ed., American

Society for Metals, Metals Park, Ohio, 1983.

1	2	3	4	5	6	7	8
		Temperature	Tensile	Yield			
AISI No.	Treatment	°C (°F)	Strength MPa (kpsi)	Strength, MPa (kpsi)	Elongation, %	Reduction in Area, %	Brinell Hardness
1030	Q&T*	205 (400)	848 (123)	648 (94)	17	47	495
	Q&T*	315 (600)	800 (116)	621 (90)	19	53	401
	Q&T*	425 (800)	731 (106)	579 (84)	23	60	302
	Q&T*	540 (1000)	669 (97)	517 (75)	28	65	255
	Q&T*	650 (1200)	586 (85)	441 (64)	32	70	207
	Normalized	925 (1700)	521 (75)	345 (50)	32	61	149
	Annealed	870 (1600)	430 (62)	317 (46)	35	64	137
1040	Q&T	205 (400)	779 (113)	593 (86)	19	48	262
	Q&T	425 (800)	758 (110)	552 (80)	21	54	241
	Q&T	650 (1200)	634 (92)	434 (63)	29	65	192
	Normalized	900 (1650)	590 (86)	374 (54)	28	55	170
	Annealed	790 (1450)	519 (75)	353 (51)	30	57	149
1050	Q&T*	205 (400)	1120 (163)	807 (117)	9	27	514
	Q&T*	425 (800)	1090 (158)	793 (115)	13	36	444
	Q&T*	650 (1200)	717 (104)	538 (78)	28	65	235
	Normalized	900 (1650)	748 (108)	427 (62)	20	39	217
	Annealed	790 (1450)	636 (92)	365 (53)	24	40	187
1060	Q&T	425 (800)	1080 (156)	765 (111)	14	41	311
	Q&T	540 (1000)	965 (140)	669 (97)	17	45	277
	Q&T	650 (1200)	800 (116)	524 (76)	23	54	229
	Normalized	900 (1650)	776 (112)	421 (61)	18	37	229
	Annealed	790 (1450)	626 (91)	372 (54)	22	38	179
1095	Q&T	315 (600)	1260 (183)	813 (118)	10	30	375
	Q&T	425 (800)	1210 (176)	772 (112)	12	32	363
	Q&T	540 (1000)	1090 (158)	676 (98)	15	37	321
	Q&T	650 (1200)	896 (130)	552 (80)	21	47	269
	Normalized	900 (1650)	1010 (147)	500 (72)	9	13	293
	Annealed	790 (1450)	658 (95)	380 (55)	13	21	192
1141	Q&T	315 (600)	1460 (212)	1280 (186)	9	32	415
	Q&T	540 (1000)	896 (130)	765 (111)	18	57	262

(continued)

Table A-21 (Continued)

Mean Mechanical Properties of Some Heat-Treated Steels

[These are typical properties for materials normalized and annealed. The properties for quenched and tempered (Q&T) steels are from a single heat. Because of the many variables, the properties listed are global averages. In all cases, data were obtained from specimens of diameter 0.505 in, machined from 1-in rounds, and of gauge length 2 in. Unless noted, all specimens were oil-quenched.] Source: *ASM Metals Reference Book*, 2d ed., American

Society for Metals, Metals Park, Ohio, 1983.

1	2	3	4	5	6	7	8
		Temperature	Tensile	Yield			
AISI No.	Treatment	°C (°F)	Strength MPa (kpsi)	Strength, MPa (kpsi)	Elongation, %	Reduction in Area, %	Brinell Hardness
4130	Q&T*	205 (400)	1630 (236)	1460 (212)	10	41	467
	Q&T*	315 (600)	1500 (217)	1380 (200)	11	43	435
	Q&T*	425 (800)	1280 (186)	1190 (173)	13	49	380
	Q&T*	540 (1000)	1030 (150)	910 (132)	17	57	315
	Q&T*	650 (1200)	814 (118)	703 (102)	22	64	245
	Normalized	870 (1600)	670 (97)	436 (63)	25	59	197
	Annealed	865 (1585)	560 (81)	361 (52)	28	56	156
4140	Q&T	205 (400)	1770 (257)	1640 (238)	8	38	510
	Q&T	315 (600)	1550 (225)	1430 (208)	9	43	445
	Q&T	425 (800)	1250 (181)	1140 (165)	13	49	370
	Q&T	540 (1000)	951 (138)	834 (121)	18	58	285
	Q&T	650 (1200)	758 (110)	655 (95)	22	63	230
	Normalized	870 (1600)	1020 (148)	655 (95)	18	47	302
	Annealed	815 (1500)	655 (95)	417 (61)	26	57	197
4340	Q&T	315 (600)	1720 (250)	1590 (230)	10	40	486
	Q&T	425 (800)	1470 (213)	1360 (198)	10	44	430
	Q&T	540 (1000)	1170 (170)	1080 (156)	13	51	360
	Q&T	650 (1200)	965 (140)	855 (124)	19	60	280

*Water-quenched

Table A-22

Results of Tensile Tests of Some Metals* Source: J. Datsko, "Solid Materials," chap. 7 in Joseph E. Shigley and Charles R. Mischke (eds.-in-chief), *Standard Handbook of Machine Design*, 2nd ed., McGraw-Hill, New York, 1996, pp. 7.47–7.50.

Number	Material	Condition	Strength (Tensile)				
			Yield		Ultimate		Strain
			S_y MPa (kpsi)	S_u MPa (kpsi)	Fracture, σ_f MPa (kpsi)	Coefficient σ_0 MPa (kpsi)	
1018	Steel	Annealed	220 (32.0)	341 (49.5)	628 (91.1) [†]	620 (90.0)	0.25
1144	Steel	Annealed	358 (52.0)	646 (93.7)	898 (130) [†]	992 (144)	0.14
1212	Steel	HR	193 (28.0)	424 (61.5)	729 (106) [†]	758 (110)	0.24
1045	Steel	Q&T 600°F	1520 (220)	1580 (230)	2380 (345)	1880 (273) [†]	0.041
4142	Steel	Q&T 600°F	1720 (250)	1930 (210)	2340 (340)	1760 (255) [†]	0.048
303	Stainless steel	Annealed	241 (35.0)	601 (87.3)	1520 (221) [†]	1410 (205)	0.51
304	Stainless steel	Annealed	276 (40.0)	568 (82.4)	1600 (233) [†]	1270 (185)	0.45
2011	Aluminum alloy	T6	169 (24.5)	324 (47.0)	325 (47.2) [†]	620 (90)	0.28
2024	Aluminum alloy	T4	296 (43.0)	446 (64.8)	533 (77.3) [†]	689 (100)	0.15
7075	Aluminum alloy	T6	542 (78.6)	593 (86.0)	706 (102) [†]	882 (128)	0.13

*Values from one or two heats and believed to be attainable using proper purchase specifications. The fracture strain may vary as much as 100 percent.

[†]Derived value.

Table A-23

Mean Monotonic and Cyclic Stress-Strain Properties of Selected Steels Source: ASM Metals Reference Book, 2nd ed., American Society for Metals, Metals Park, Ohio, 1983, p. 217.

Grade (a)	Orienta- tion (e)	Description (f)	Hard- ness		Tensile Strength S _{ut}		Reduction in Area %	True Strain at Fracture ε _f	Modulus of Elasticity E		Fatigue Strength Coefficient σ' _f		Fatigue Strength Exponent b	Fatigue Ductility Coefficient ε' _f	Fatigue Ductility Exponent c
			HB		MPa	ksi			GPa	10 ⁴ psi	MPa	ksi			
A538A (b)	L	STA	405		1515	220	67	1.10	185	27	1655	240	-0.065	0.30	-0.62
A538B (b)	L	STA	460		1860	270	56	0.82	185	27	2135	310	-0.071	0.80	-0.71
A538C (b)	L	STA	480		2000	290	55	0.81	180	26	2240	325	-0.07	0.60	-0.75
AM-350 (c)	L	HR, A			1315	191	52	0.74	195	28	2800	406	-0.14	0.33	-0.84
AM-350 (c)	L	CD	496		1905	276	20	0.23	180	26	2690	390	-0.102	0.10	-0.42
Gainex (c)	LT	HR sheet			530	77	58	0.86	200	29.2	805	117	-0.07	0.86	-0.65
Gainex (c)	L	HR sheet			510	74	64	1.02	200	29.2	805	117	-0.071	0.86	-0.68
H-11	L	Ausformed	660		2585	375	33	0.40	205	30	3170	460	-0.077	0.08	-0.74
RQC-100 (c)	LT	HR plate	290		940	136	43	0.56	205	30	1240	180	-0.07	0.66	-0.69
RQC-100 (c)	L	HR plate	290		930	135	67	1.02	205	30	1240	180	-0.07	0.66	-0.69
10B62	L	Q&T	430		1640	238	38	0.89	195	28	1780	258	-0.067	0.32	-0.56
1005-1009	LT	HR sheet	90		360	52	73	1.3	205	30	580	84	-0.09	0.15	-0.43
1005-1009	LT	CD sheet	125		470	68	66	1.09	205	30	515	75	-0.059	0.30	-0.51
1005-1009	L	CD sheet	125		415	60	64	1.02	200	29	540	78	-0.073	0.11	-0.41
1005-1009	L	HR sheet	90		345	50	80	1.6	200	29	640	93	-0.109	0.10	-0.39
1015	L	Normalized	80		415	60	68	1.14	205	30	825	120	-0.11	0.95	-0.64
1020	L	HR plate	108		440	64	62	0.96	205	29.5	895	130	-0.12	0.41	-0.51
1040	L	As forged	225		620	90	60	0.93	200	29	1540	223	-0.14	0.61	-0.57
1045	L	Q&T	225		725	105	65	1.04	200	29	1225	178	-0.095	1.00	-0.66
1045	L	Q&T	410		1450	210	51	0.72	200	29	1860	270	-0.073	0.60	-0.70
1045	L	Q&T	390		1345	195	59	0.89	205	30	1585	230	-0.074	0.45	-0.68
1045	L	Q&T	450		1585	230	55	0.81	205	30	1795	260	-0.07	0.35	-0.69
1045	L	Q&T	500		1825	265	51	0.71	205	30	2275	330	-0.08	0.25	-0.68
1045	L	Q&T	595		2240	325	41	0.52	205	30	2725	395	-0.081	0.07	-0.60
1144	L	CDSR	265		930	135	33	0.51	195	28.5	1000	145	-0.08	0.32	-0.58

1144	L	DAI	305	1035	150	25	0.29	200	28.8	1585	230	-0.09	0.27	-0.53
1541F	L	Q&T forging	290	950	138	49	0.68	205	29.9	1275	185	-0.076	0.68	-0.65
1541F	L	Q&T forging	260	890	129	60	0.93	205	29.9	1275	185	-0.071	0.93	-0.65
4130	L	Q&T	258	895	130	67	1.12	220	32	1275	185	-0.083	0.92	-0.63
4130	L	Q&T	365	1425	207	55	0.79	200	29	1695	246	-0.081	0.89	-0.69
4140	L	Q&T, DAI	310	1075	156	60	0.69	200	29.2	1825	265	-0.08	1.2	-0.59
4142	L	DAI	310	1060	154	29	0.35	200	29	1450	210	-0.10	0.22	-0.51
4142	L	DAI	335	1250	181	28	0.34	200	28.9	1250	181	-0.08	0.06	-0.62
4142	L	Q&T	380	1415	205	48	0.66	205	30	1825	265	-0.08	0.45	-0.75
4142	L	Q&T and deformed	400	1550	225	47	0.63	200	29	1895	275	-0.09	0.50	-0.75
4142	L	Q&T	450	1760	255	42	0.54	205	30	2000	290	-0.08	0.40	-0.73
4142	L	Q&T and deformed	475	2035	295	20	0.22	200	29	2070	300	-0.082	0.20	-0.77
4142	L	Q&T and deformed	450	1930	280	37	0.46	200	29	2105	305	-0.09	0.60	-0.76
4142	L	Q&T	475	1930	280	35	0.43	205	30	2170	315	-0.081	0.09	-0.61
4142	L	Q&T	560	2240	325	27	0.31	205	30	2655	385	-0.089	0.07	-0.76
4340	L	HR, A	243	825	120	43	0.57	195	28	1200	174	-0.095	0.45	-0.54
4340	L	Q&T	409	1470	213	38	0.48	200	29	2000	290	-0.091	0.48	-0.60
4340	L	Q&T	350	1240	180	57	0.84	195	28	1655	240	-0.076	0.73	-0.62
5160	L	Q&T	430	1670	242	42	0.87	195	28	1930	280	-0.071	0.40	-0.57
52100	L	SH, Q&T	518	2015	292	11	0.12	205	30	2585	375	-0.09	0.18	-0.56
9262	L	A	260	925	134	14	0.16	205	30	1040	151	-0.071	0.16	-0.47
9262	L	Q&T	280	1000	145	33	0.41	195	28	1220	177	-0.073	0.41	-0.60
9262	L	Q&T	410	565	227	32	0.38	200	29	1855	269	-0.057	0.38	-0.65
950C (d)	LT	HR plate	159	565	82	64	1.03	205	29.6	1170	170	-0.12	0.95	-0.61
950C (d)	L	HR bar	150	565	82	69	1.19	205	30	970	141	-0.11	0.85	-0.59
950X (d)	L	Plate channel	150	440	64	65	1.06	205	30	625	91	-0.075	0.35	-0.54
950X (d)	L	HR plate	156	530	77	72	1.24	205	29.5	1005	146	-0.10	0.85	-0.61
950X (d)	L	Plate channel	225	695	101	68	1.15	195	28.2	1055	153	-0.08	0.21	-0.53

Notes: (a) AISI/SAE grade, unless otherwise indicated. (b) ASTM designation. (c) Proprietary designation. (d) SAE HSLA grade. (e) Orientation of axis of specimen, relative to rolling direction; L is longitudinal (parallel to rolling direction); LT is long transverse (perpendicular to rolling direction). (f) STA, solution treated and aged; HR, hot rolled; CD, cold drawn; Q&T, quenched and tempered; CDSR, cold drawn strain relieved; DAT, drawn at temperature; A, annealed.

From *ASM Metals Reference Book, 2nd edition*, 1983; ASM International, Materials Park, OH 44073-0002; table 217. Reprinted by permission of ASM International®, www.asminternational.org.

Table A-24

Mechanical Properties of Three Non-Steel Metals

(a) Typical Properties of Gray Cast Iron

[The American Society for Testing and Materials (ASTM) numbering system for gray cast iron is such that the numbers correspond to the *minimum tensile strength* in kpsi. Thus an ASTM No. 20 cast iron has a minimum tensile strength of 20 kpsi. Note particularly that the tabulations are *typical* of several heats.]

ASTM Number	Tensile Strength S_{UT} , kpsi	Compressive Strength S_{UC} , kpsi	Shear Modulus of Rupture S_{SR} , kpsi	Modulus of Elasticity, Mpsi		Endurance Limit*, S_e , kpsi	Brinell Hardness H_B	Fatigue Stress- Concentration Factor K_f	
				Tension [†]	Torsion				
20	22	83	26	9.6–14	3.9–5.6	10	156	1.00	
25	26	97	32	11.5–14.8	4.6–6.0	11.5	174	1.05	
30	31	109	40	13–16.4	5.2–6.6	14	201	1.10	
35	36.5	124	48.5	14.5–17.2	5.8–6.9	16	212	1.15	
40	42.5	140	57	16–20	6.4–7.8	18.5	235	1.25	
50	52.5	164	73	18.8–22.8	7.2–8.0	21.5	262	1.35	
60	62.5	187.5	88.5	20.4–23.5	7.8–8.5	24.5	302	1.50	

*Polished or machined specimens.

[†]The modulus of elasticity of cast iron in compression corresponds closely to the upper value in the range given for tension and is a more constant value than that for tension.

Table A-24Mechanical Properties of Three Non-Steel Metals (*Continued*)

(b) Mechanical Properties of Some Aluminum Alloys

[These are *typical* properties for sizes of about $\frac{1}{2}$ in; similar properties can be obtained by using proper purchase specifications. The values given for fatigue strength correspond to $50(10^7)$ cycles of completely reversed stress. Aluminum alloys do not have an endurance limit. Yield strengths were obtained by the 0.2 percent offset method.]

Aluminum Association Number	Temper	Strength			Elongation in 2 in, %	Brinell Hardness H_B
		Yield, S_y , MPa (kpsi)	Tensile, S_{UT} , MPa (kpsi)	Fatigue, S_f , MPa (kpsi)		
Wrought:						
2017	O	70 (10)	179 (26)	90 (13)	22	45
2024	O	76 (11)	186 (27)	90 (13)	22	47
	T3	345 (50)	482 (70)	138 (20)	16	120
3003	H12	117 (17)	131 (19)	55 (8)	20	35
	H16	165 (24)	179 (26)	65 (9.5)	14	47
3004	H34	186 (27)	234 (34)	103 (15)	12	63
	H38	234 (34)	276 (40)	110 (16)	6	77
5052	H32	186 (27)	234 (34)	117 (17)	18	62
	H36	234 (34)	269 (39)	124 (18)	10	74
Cast:						
319.0*	T6	165 (24)	248 (36)	69 (10)	2.0	80
333.0†	T5	172 (25)	234 (34)	83 (12)	1.0	100
	T6	207 (30)	289 (42)	103 (15)	1.5	105
335.0*	T6	172 (25)	241 (35)	62 (9)	3.0	80
	T7	248 (36)	262 (38)	62 (9)	0.5	85

*Sand casting.

†Permanent-mold casting.

(c) Mechanical Properties of Some Titanium Alloys

Titanium Alloy	Condition	Yield, S_y (0.2% offset) MPa (kpsi)	Strength Tensile, S_{ut} MPa (kpsi)	Elongation in 2 in, %	Hardness (Brinell or Rockwell)
Ti-35A†	Annealed	210 (30)	275 (40)	30	135 HB
Ti-50A†	Annealed	310 (45)	380 (55)	25	215 HB
Ti-0.2 Pd	Annealed	280 (40)	340 (50)	28	200 HB
Ti-5 Al-2.5 Sn	Annealed	760 (110)	790 (115)	16	36 HRC
Ti-8 Al-1 Mo-1 V	Annealed	900 (130)	965 (140)	15	39 HRC
Ti-6 Al-6 V-2 Sn	Annealed	970 (140)	1030 (150)	14	38 HRC
Ti-6Al-4V	Annealed	900 (130)	830 (120)	14	36 HRC
Ti-13 V-11 Cr-3 Al	Sol. + aging	1207 (175)	1276 (185)	8	40 HRC

†Commercially pure alpha titanium

Table A-26

Stochastic Parameters for Finite Life Fatigue Tests in Selected Metals Source: E. B. Haugen, *Probabilistic Mechanical Design*, Wiley, New York, 1980, Appendix 10-B.

1	2	3	4	5	6	7	8	9
Number	Condition	T_S MPa (kpsi)	Y_S MPa (kpsi)	Distribution	10^4	10^5	10^6	10^7
1046	WQ&T, 1210°F	723 (105)	565 (82)	W x_0	544 (79)	462 (67)	391 (56.7)	
				θ	594 (86.2)	503 (73.0)	425 (61.7)	
				b	2.60	2.75	2.85	
2340	OQ&T 1200°F	799 (116)	661 (96)	W x_0	579 (84)	510 (74)	420 (61)	
				θ	699 (101.5)	588 (85.4)	496 (72.0)	
				b	4.3	3.4	4.1	
3140	OQ&T, 1300°F	744 (108)	599 (87)	W x_0	510 (74)	455 (66)	393 (57)	
				θ	604 (87.7)	528 (76.7)	463 (67.2)	
				b	5.2	5.0	5.5	
2024 Aluminum	T-4	489 (71)	365 (53)	N σ	26.3 (3.82)	21.4 (3.11)	17.4 (2.53)	14.0 (2.03)
				μ	143 (20.7)	116 (16.9)	95 (13.8)	77 (11.2)
Ti-6Al-4V	HT-46	1040 (151)	992 (144)	N σ	39.6 (5.75)	38.1 (5.53)	36.6 (5.31)	35.1 (5.10)
				μ	712 (108)	684 (99.3)	657 (95.4)	493 (71.6)

Statistical parameters from a large number of fatigue tests are listed. Weibull distribution is denoted W and the parameters are x_0 , "guaranteed" fatigue strength; θ , characteristic fatigue strength; and b , shape factor. Normal distribution is denoted N and the parameters are μ , mean fatigue strength, and σ , standard deviation of the fatigue strength. The life is in stress-cycles-to-failure. T_S = tensile strength, Y_S = yield strength. All testing by rotating-beam specimen.

Table A-27

Finite Life Fatigue Strengths of Selected Plain Carbon Steels Source: Compiled from Table 4 in H. J. Grover, S. A. Gordon, and L. R. Jackson, *Fatigue of Metals and Structures*, Bureau of Naval Weapons Document NAVWEPS 00-25-534, 1960.

Material	Condition	BHN*	Tensile		RA*	Stress Cycles to Failure							
			Strength kpsi	Yield Strength kpsi		10 ⁴	4(10 ⁴)	10 ⁵	4(10 ⁵)	10 ⁶	4(10 ⁶)	10 ⁷	10 ⁸
1020	Furnace cooled		58	30	0.63			37	34	30	28	25	
1030	Air-cooled	135	80	45	0.62		51	47	42	38	38	38	
1035	Normal	132	72	35	0.54			44	40	37	34	33	33
	WQT	209	103	87	0.65		80	72	65	60	57	57	57
1040	Forged	195	92	53	0.23				40	47	33	33	
1045	HR, N		107	63	0.49	80	70	56	47	47	47	47	
1050	N, AC	164	92	47	0.40	50	48	46	40	38	34	34	
	WQT												
	1200	196	97	70	0.58		60	57	52	50	50	50	50
.56 MN	N	193	98	47	0.42	61	55	51	47	43	41	41	41
	WQT	277	111	84	0.57	94	81	73	62	57	55	55	55
	1200												
1060	As Rec.	67 Rb	134	65	0.20	65	60	55	50	48	48	48	
1095		162	84	33	0.37	50	43	40	34	31	30	30	30
	OQT	227	115	65	0.40	77	68	64	57	56	56	56	56
	1200												
10120		224	117	59	0.12		60	56	51	50	50	50	
	OQT	369	180	130	0.15		102	95	91	91	91	91	91
	860												

*BHN = Brinell hardness number, RA = fractional reduction in area.

Table A-28

Decimal Equivalents of Wire and Sheet/Metal Gauges* (All Sizes Are Given in Inches)

Name of Gauge:	American or Brown & Sharpe	Birmingham or Stubs Iron Wire	United States Standard†	Manu- facturers Standard	Steel Wire or Washburn & Moen		Twist Drills and Drill Steel
					Music Wire	Stubs Steel Wire	
Principal Uses:	Nonferrous Sheet, Wire, and Rod	Tubing, Ferrous Strip, Flat Wire, and Spring Steel	Ferrous Sheet and Plate, 480 lbf/ft ³	Ferrous Sheet	Ferrous Music Wire	Steel Drill Rod	Twist Drills and Drill Steel
7/0			0.500		0.490		
6/0	0.580 0		0.468 75		0.461 5		
5/0	0.516 5		0.437 5		0.430 5		
4/0	0.460 0	0.454	0.406 25		0.393 8		
3/0	0.409 6	0.425	0.375		0.362 5		
2/0	0.364 8	0.380	0.343 75		0.331 0		
0	0.324 9	0.340	0.312 5		0.306 5		
1	0.289 3	0.300	0.281 25		0.283 0		0.227
2	0.257 6	0.284	0.265 625		0.262 5		0.219
3	0.229 4	0.259	0.25	0.239 1	0.243 7		0.212
4	0.204 3	0.238	0.234 375	0.224 2	0.225 3		0.207
5	0.181 9	0.220	0.218 75	0.209 2	0.207 0		0.204
6	0.162 0	0.203	0.203 125	0.194 3	0.192 0		0.201
7	0.144 3	0.180	0.187 5	0.179 3	0.177 0		0.199
8	0.128 5	0.165	0.171 875	0.164 4	0.162 0		0.197
9	0.114 4	0.148	0.156 25	0.149 5	0.148 3		0.194
10	0.101 9	0.134	0.140 625	0.134 5	0.135 0		0.191
11	0.090 74	0.120	0.125	0.119 6	0.120 5		0.188
12	0.080 81	0.109	0.109 357	0.104 6	0.105 5		0.185
13	0.071 96	0.095	0.093 75	0.089 7	0.091 5		0.182
14	0.064 08	0.083	0.078 125	0.074 7	0.080 0		0.180
15	0.057 07	0.072	0.070 312 5	0.067 3	0.072 0		0.178
16	0.050 82	0.065	0.062 5	0.059 8	0.062 5		0.175
17	0.045 26	0.058	0.056 25	0.053 8	0.054 0		0.172

(continued)

Table A-28

Decimal Equivalents of Wire and Sheet/Metal Gauges* (All Sizes Are Given in Inches) (Continued)

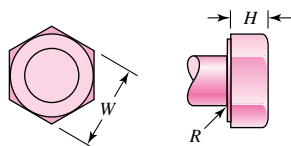
Name of Gauge:	American or Brown & Sharpe	Birmingham or Stubs Iron Wire	United States Standard†	Manu- facturers Standard	Steel Wire		Twist Drills and Drill Steel
					or Washburn & Moen	Music Wire	
Principal Use:	Nonferrous Sheet, Wire, and Rod	Tubing, Ferrous Strip, Flat Wire, and Spring Steel	Ferrous Sheet and Plate, 480 lbf/ft³	Ferrous Sheet	Ferrous Wire Except Music Wire	Music Wire	Steel Drill Rod
18	0.040 30	0.049	0.05	0.047 8	0.047 5	0.041	0.168
19	0.035 89	0.042	0.043 75	0.041 8	0.041 0	0.043	0.164
20	0.031 96	0.035	0.037 5	0.035 9	0.034 8	0.045	0.161
21	0.028 46	0.032	0.034 375	0.032 9	0.031 7	0.047	0.157
22	0.025 35	0.028	0.031 25	0.029 9	0.028 6	0.049	0.155
23	0.022 57	0.025	0.028 125	0.026 9	0.025 8	0.051	0.153
24	0.020 10	0.022	0.025	0.023 9	0.023 0	0.055	0.151
25	0.017 90	0.020	0.021 875	0.020 9	0.020 4	0.059	0.148
26	0.015 94	0.018	0.018 75	0.017 9	0.018 1	0.063	0.146
27	0.014 20	0.016	0.017 187 5	0.016 4	0.017 3	0.067	0.143
28	0.012 64	0.014	0.015 625	0.014 9	0.016 2	0.071	0.139
29	0.011 26	0.013	0.014 062 5	0.013 5	0.015 0	0.075	0.134
30	0.010 03	0.012	0.012 5	0.012 0	0.014 0	0.080	0.127
31	0.008 928	0.010	0.010 937 5	0.010 5	0.013 2	0.085	0.120
32	0.007 950	0.009	0.010 156 25	0.009 7	0.012 8	0.090	0.115
33	0.007 080	0.008	0.009 375	0.009 0	0.011 8	0.095	0.112
34	0.006 305	0.007	0.008 593 75	0.008 2	0.010 4		0.110
35	0.005 615	0.005	0.007 812 5	0.007 5	0.009 5		0.108
36	0.005 000	0.004	0.007 031 25	0.006 7	0.009 0		0.106
37	0.004 453		0.006 640 625	0.006 4	0.008 5		0.103
38	0.003 965		0.006 25	0.006 0	0.008 0		0.101
39	0.003 531				0.007 5		0.099
40	0.003 145				0.007 0		0.097

*Specify sheet, wire, and plate by stating the gauge number, the gauge name, and the decimal equivalent in parentheses.

†Reflects present average and weights of sheet steel.

Table A-29

Dimensions of Square and Hexagonal Bolts



Head Type											
Nominal Size, in	Square		Regular Hexagonal			Heavy Hexagonal			Structural Hexagonal		
	W	H	W	H	R _{min}	W	H	R _{min}	W	H	R _{min}
$\frac{1}{4}$	$\frac{3}{8}$	$\frac{11}{64}$	$\frac{7}{16}$	$\frac{11}{64}$	0.01						
$\frac{5}{16}$	$\frac{1}{2}$	$\frac{13}{64}$	$\frac{1}{2}$	$\frac{7}{32}$	0.01						
$\frac{3}{8}$	$\frac{9}{16}$	$\frac{1}{4}$	$\frac{9}{16}$	$\frac{1}{4}$	0.01						
$\frac{7}{16}$	$\frac{5}{8}$	$\frac{19}{64}$	$\frac{5}{8}$	$\frac{19}{64}$	0.01						
$\frac{1}{2}$	$\frac{3}{4}$	$\frac{21}{64}$	$\frac{3}{4}$	$\frac{11}{32}$	0.01	$\frac{7}{8}$	$\frac{11}{32}$	0.01	$\frac{7}{8}$	$\frac{5}{16}$	0.009
$\frac{5}{8}$	$\frac{15}{16}$	$\frac{27}{64}$	$\frac{15}{16}$	$\frac{27}{64}$	0.02	$1\frac{1}{16}$	$\frac{27}{64}$	0.02	$1\frac{1}{16}$	$\frac{25}{64}$	0.021
$\frac{3}{4}$	$1\frac{1}{8}$	$\frac{1}{2}$	$1\frac{1}{8}$	$\frac{1}{2}$	0.02	$1\frac{1}{4}$	$\frac{1}{2}$	0.02	$1\frac{1}{4}$	$\frac{15}{32}$	0.021
1	$1\frac{1}{2}$	$\frac{21}{32}$	$1\frac{1}{2}$	$\frac{43}{64}$	0.03	$1\frac{5}{8}$	$\frac{43}{64}$	0.03	$1\frac{5}{8}$	$\frac{39}{64}$	0.062
$1\frac{1}{8}$	$1\frac{11}{16}$	$\frac{3}{4}$	$1\frac{11}{16}$	$\frac{3}{4}$	0.03	$1\frac{13}{16}$	$\frac{3}{4}$	0.03	$1\frac{13}{16}$	$\frac{11}{16}$	0.062
$1\frac{1}{4}$	$1\frac{7}{8}$	$\frac{27}{32}$	$1\frac{7}{8}$	$\frac{27}{32}$	0.03	2	$\frac{27}{32}$	0.03	2	$\frac{25}{32}$	0.062
$1\frac{3}{8}$	$2\frac{1}{16}$	$\frac{29}{32}$	$2\frac{1}{16}$	$\frac{29}{32}$	0.03	$2\frac{3}{16}$	$\frac{29}{32}$	0.03	$2\frac{3}{16}$	$\frac{27}{32}$	0.062
$1\frac{1}{2}$	$2\frac{1}{4}$	1	$2\frac{1}{4}$	1	0.03	$2\frac{3}{8}$	1	0.03	$2\frac{3}{8}$	$\frac{15}{16}$	0.062
Nominal Size, mm											
M5	8	3.58	8	3.58	0.2						
M6			10	4.38	0.3						
M8			13	5.68	0.4						
M10			16	6.85	0.4						
M12			18	7.95	0.6	21	7.95	0.6			
M14			21	9.25	0.6	24	9.25	0.6			
M16			24	10.75	0.6	27	10.75	0.6	27	10.75	0.6
M20			30	13.40	0.8	34	13.40	0.8	34	13.40	0.8
M24			36	15.90	0.8	41	15.90	0.8	41	15.90	1.0
M30			46	19.75	1.0	50	19.75	1.0	50	19.75	1.2
M36			55	23.55	1.0	60	23.55	1.0	60	23.55	1.5

Table A-30

Dimensions of Hexagonal Cap Screws and Heavy Hexagonal Screws (W = Width across Flats; H = Height of Head; See Figure in Table A-29)

Nominal Size, in	Minimum Fillet Radius	Type of Screw		Height H
		Cap W	Heavy W	
$\frac{1}{4}$	0.015	$\frac{7}{16}$		$\frac{5}{32}$
$\frac{5}{16}$	0.015	$\frac{1}{2}$		$\frac{13}{64}$
$\frac{3}{8}$	0.015	$\frac{9}{16}$		$\frac{15}{64}$
$\frac{7}{16}$	0.015	$\frac{5}{8}$		$\frac{9}{32}$
$\frac{1}{2}$	0.015	$\frac{3}{4}$	$\frac{7}{8}$	$\frac{5}{16}$
$\frac{5}{8}$	0.020	$\frac{15}{16}$	$1\frac{1}{16}$	$\frac{25}{64}$
$\frac{3}{4}$	0.020	$1\frac{1}{8}$	$1\frac{1}{4}$	$\frac{15}{32}$
$\frac{7}{8}$	0.040	$1\frac{5}{16}$	$1\frac{7}{16}$	$\frac{35}{64}$
1	0.060	$1\frac{1}{2}$	$1\frac{1}{8}$	$\frac{39}{64}$
$1\frac{1}{4}$	0.060	$1\frac{7}{8}$	2	$\frac{25}{32}$
$1\frac{3}{8}$	0.060	$2\frac{1}{16}$	$2\frac{3}{16}$	$\frac{27}{32}$
$1\frac{1}{2}$	0.060	$2\frac{1}{4}$	$2\frac{3}{8}$	$\frac{15}{16}$
Nominal Size, mm				
M5	0.2	8		3.65
M6	0.3	10		4.15
M8	0.4	13		5.50
M10	0.4	16		6.63
M12	0.6	18	21	7.76
M14	0.6	21	24	9.09
M16	0.6	24	27	10.32
M20	0.8	30	34	12.88
M24	0.8	36	41	15.44
M30	1.0	46	50	19.48
M36	1.0	55	60	23.38

Table A-31

Dimensions of
Hexagonal Nuts

Nominal Size, in	Width W	Height H		
		Regular Hexagonal	Thick or Slotted	JAM
$\frac{1}{4}$	$\frac{7}{16}$	$\frac{7}{32}$	$\frac{9}{32}$	$\frac{5}{32}$
$\frac{5}{16}$	$\frac{1}{2}$	$\frac{17}{64}$	$\frac{21}{64}$	$\frac{3}{16}$
$\frac{3}{8}$	$\frac{9}{16}$	$\frac{21}{64}$	$\frac{13}{32}$	$\frac{7}{32}$
$\frac{7}{16}$	$\frac{11}{16}$	$\frac{3}{8}$	$\frac{29}{64}$	$\frac{1}{4}$
$\frac{1}{2}$	$\frac{3}{4}$	$\frac{7}{16}$	$\frac{9}{16}$	$\frac{5}{16}$
$\frac{9}{16}$	$\frac{7}{8}$	$\frac{31}{64}$	$\frac{39}{64}$	$\frac{5}{16}$
$\frac{5}{8}$	$\frac{15}{16}$	$\frac{35}{64}$	$\frac{23}{32}$	$\frac{3}{8}$
$\frac{3}{4}$	$1\frac{1}{8}$	$\frac{41}{64}$	$\frac{13}{16}$	$\frac{27}{64}$
$\frac{7}{8}$	$1\frac{5}{16}$	$\frac{3}{4}$	$\frac{29}{32}$	$\frac{31}{64}$
1	$1\frac{1}{2}$	$\frac{55}{64}$	1	$\frac{35}{64}$
$1\frac{1}{8}$	$1\frac{11}{16}$	$\frac{31}{32}$	$1\frac{5}{32}$	$\frac{39}{64}$
$1\frac{1}{4}$	$1\frac{7}{8}$	$1\frac{1}{16}$	$1\frac{1}{4}$	$\frac{23}{32}$
$1\frac{3}{8}$	$2\frac{1}{16}$	$1\frac{11}{64}$	$1\frac{3}{8}$	$\frac{25}{32}$
$1\frac{1}{2}$	$2\frac{1}{4}$	$1\frac{9}{32}$	$1\frac{1}{2}$	$\frac{27}{32}$
Nominal Size, mm				
M5	8	4.7	5.1	2.7
M6	10	5.2	5.7	3.2
M8	13	6.8	7.5	4.0
M10	16	8.4	9.3	5.0
M12	18	10.8	12.0	6.0
M14	21	12.8	14.1	7.0
M16	24	14.8	16.4	8.0
M20	30	18.0	20.3	10.0
M24	36	21.5	23.9	12.0
M30	46	25.6	28.6	15.0
M36	55	31.0	34.7	18.0

Table A-32

Basic Dimensions of
American Standard
Plain Washers (All
Dimensions in Inches)

Fastener Size	Washer Size	Diameter		Thickness
		ID	OD	
#6	0.138	0.156	0.375	0.049
#8	0.164	0.188	0.438	0.049
#10	0.190	0.219	0.500	0.049
$\frac{3}{16}$	0.188	0.250	0.562	0.049
#12	0.216	0.250	0.562	0.065
$\frac{1}{4}$ N	0.250	0.281	0.625	0.065
$\frac{1}{4}$ W	0.250	0.312	0.734	0.065
$\frac{5}{16}$ N	0.312	0.344	0.688	0.065
$\frac{5}{16}$ W	0.312	0.375	0.875	0.083
$\frac{3}{8}$ N	0.375	0.406	0.812	0.065
$\frac{3}{8}$ W	0.375	0.438	1.000	0.083
$\frac{7}{16}$ N	0.438	0.469	0.922	0.065
$\frac{7}{16}$ W	0.438	0.500	1.250	0.083
$\frac{1}{2}$ N	0.500	0.531	1.062	0.095
$\frac{1}{2}$ W	0.500	0.562	1.375	0.109
$\frac{9}{16}$ N	0.562	0.594	1.156	0.095
$\frac{9}{16}$ W	0.562	0.625	1.469	0.109
$\frac{5}{8}$ N	0.625	0.656	1.312	0.095
$\frac{5}{8}$ W	0.625	0.688	1.750	0.134
$\frac{3}{4}$ N	0.750	0.812	1.469	0.134
$\frac{3}{4}$ W	0.750	0.812	2.000	0.148
$\frac{7}{8}$ N	0.875	0.938	1.750	0.134
$\frac{7}{8}$ W	0.875	0.938	2.250	0.165
1 N	1.000	1.062	2.000	0.134
1 W	1.000	1.062	2.500	0.165
$1\frac{1}{8}$ N	1.125	1.250	2.250	0.134
$1\frac{1}{8}$ W	1.125	1.250	2.750	0.165
$1\frac{1}{4}$ N	1.250	1.375	2.500	0.165
$1\frac{1}{4}$ W	1.250	1.375	3.000	0.165
$1\frac{3}{8}$ N	1.375	1.500	2.750	0.165
$1\frac{3}{8}$ W	1.375	1.500	3.250	0.180
$1\frac{1}{2}$ N	1.500	1.625	3.000	0.165
$1\frac{1}{2}$ W	1.500	1.625	3.500	0.180
$1\frac{5}{8}$	1.625	1.750	3.750	0.180
$1\frac{3}{4}$	1.750	1.875	4.000	0.180
$1\frac{7}{8}$	1.875	2.000	4.250	0.180
2	2.000	2.125	4.500	0.180
$2\frac{1}{4}$	2.250	2.375	4.750	0.220
$2\frac{1}{2}$	2.500	2.625	5.000	0.238
$2\frac{3}{4}$	2.750	2.875	5.250	0.259
3	3.000	3.125	5.500	0.284

N = narrow; W = wide; use W when not specified.

Table A-33

Dimensions of Metric Plain Washers (All Dimensions in Millimeters)

Washer Size*	Minimum ID	Maximum OD	Maximum Thickness	Washer Size*	Minimum ID	Maximum OD	Maximum Thickness
1.6 N	1.95	4.00	0.70	10 N	10.85	20.00	2.30
1.6 R	1.95	5.00	0.70	10 R	10.85	28.00	2.80
1.6 W	1.95	6.00	0.90	10 W	10.85	39.00	3.50
2 N	2.50	5.00	0.90	12 N	13.30	25.40	2.80
2 R	2.50	6.00	0.90	12 R	13.30	34.00	3.50
2 W	2.50	8.00	0.90	12 W	13.30	44.00	3.50
2.5 N	3.00	6.00	0.90	14 N	15.25	28.00	2.80
2.5 R	3.00	8.00	0.90	14 R	15.25	39.00	3.50
2.5 W	3.00	10.00	1.20	14 W	15.25	50.00	4.00
3 N	3.50	7.00	0.90	16 N	17.25	32.00	3.50
3 R	3.50	10.00	1.20	16 R	17.25	44.00	4.00
3 W	3.50	12.00	1.40	16 W	17.25	56.00	4.60
3.5 N	4.00	9.00	1.20	20 N	21.80	39.00	4.00
3.5 R	4.00	10.00	1.40	20 R	21.80	50.00	4.60
3.5 W	4.00	15.00	1.75	20 W	21.80	66.00	5.10
4 N	4.70	10.00	1.20	24 N	25.60	44.00	4.60
4 R	4.70	12.00	1.40	24 R	25.60	56.00	5.10
4 W	4.70	16.00	2.30	24 W	25.60	72.00	5.60
5 N	5.50	11.00	1.40	30 N	32.40	56.00	5.10
5 R	5.50	15.00	1.75	30 R	32.40	72.00	5.60
5 W	5.50	20.00	2.30	30 W	32.40	90.00	6.40
6 N	6.65	13.00	1.75	36 N	38.30	66.00	5.60
6 R	6.65	18.80	1.75	36 R	38.30	90.00	6.40
6 W	6.65	25.40	2.30	36 W	38.30	110.00	8.50
8 N	8.90	18.80	2.30				
8 R	8.90	25.40	2.30				
8 W	8.90	32.00	2.80				

N = narrow; R = regular; W = wide.

*Same as screw or bolt size.

Table A-34

Gamma Function*

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$$\text{Values of } \Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx; \Gamma(n+1) = n\Gamma(n)$$

n	$\Gamma(n)$	n	$\Gamma(n)$	n	$\Gamma(n)$	n	$\Gamma(n)$
1.00	1.000 00	1.25	.906 40	1.50	.886 23	1.75	.919 06
1.01	.994 33	1.26	.904 40	1.51	.886 59	1.76	.921 37
1.02	.988 84	1.27	.902 50	1.52	.887 04	1.77	.923 76
1.03	.983 55	1.28	.900 72	1.53	.887 57	1.78	.926 23
1.04	.978 44	1.29	.899 04	1.54	.888 18	1.79	.928 77
1.05	.973 50	1.30	.897 47	1.55	.888 87	1.80	.931 38
1.06	.968 74	1.31	.896 00	1.56	.889 64	1.81	.934 08
1.07	.964 15	1.32	.894 64	1.57	.890 49	1.82	.936 85
1.08	.959 73	1.33	.893 38	1.58	.891 42	1.83	.939 69
1.09	.955 46	1.34	.892 22	1.59	.892 43	1.84	.942 61
1.10	.951 35	1.35	.891 15	1.60	.893 52	1.85	.945 61
1.11	.947 39	1.36	.890 18	1.61	.894 68	1.86	.948 69
1.12	.943 59	1.37	.889 31	1.62	.895 92	1.87	.951 84
1.13	.939 93	1.38	.888 54	1.63	.897 24	1.88	.955 07
1.14	.936 42	1.39	.887 85	1.64	.898 64	1.89	.958 38
1.15	.933 04	1.40	.887 26	1.65	.900 12	1.90	.961 77
1.16	.929 80	1.41	.886 76	1.66	.901 67	1.91	.965 23
1.17	.926 70	1.42	.886 36	1.67	.903 30	1.92	.968 78
1.18	.923 73	1.43	.886 04	1.68	.905 00	1.93	.972 40
1.19	.920 88	1.44	.885 80	1.69	.906 78	1.94	.976 10
1.20	.918 17	1.45	.885 65	1.70	.908 64	1.95	.979 88
1.21	.915 58	1.46	.885 60	1.71	.910 57	1.96	.983 74
1.22	.913 11	1.47	.885 63	1.72	.912 58	1.97	.987 68
1.23	.910 75	1.48	.885 75	1.73	.914 66	1.98	.991 71
1.24	.908 52	1.49	.885 95	1.74	.916 83	1.99	.995 81
						2.00	1.000 00

*For large positive values of x , $\Gamma(x)$ approximates the asymptotic series

$$x^x e^{-x} \sqrt{\frac{2x}{x}} \left[1 + \frac{1}{12x} + \frac{1}{288x^2} - \frac{139}{51\,840x^3} - \frac{571}{2\,488\,320x^4} + \cdots \right]$$